

**MARKET RETAIL ANALYTICS
PROJECT-ML1**

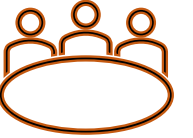
**AUTOMOBILE PARTS CUSTOMER
TRANSACTION ANALYSIS**

RESHMA B

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AGENDA



- Problem Statement
- Key Objectives
- Data Overview
- Exploratory Data Analysis (EDA)
- Observations and Insights from EDA
- Customer Segmentation using RFM analysis
- KNIME workflow and Output table
- Targeted Marketing Strategies

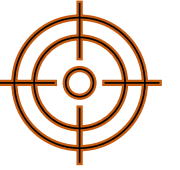


As a result, the company is unable to fully understand customer behavior, optimize customer segmentation, or implement targeted marketing strategies. This lack of data-driven insights limits the company's ability to improve customer satisfaction and retain customers.

By doing so, the company can drive revenue growth, improve service efficiency, and offer more tailored solutions to its customers.



KEY OBJECTIVES

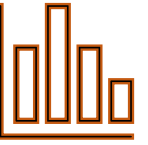


- Identifying patterns and trends in customer purchasing behavior to uncover insights into what drives customer decisions
- Segmenting customers based on transactional data to tailor marketing efforts for different customer groups
- Providing actionable insights to enhance the company's marketing strategies, driving better targeting and improving customer satisfaction
- Recommending personalized marketing strategies for each customer segment, aimed at increasing sales and fostering long-term customer loyalty

The goal is to understand customer behavior through Customer Segmentation using RFM Analysis (Recency, Frequency, Monetary) which will identify best customers, loyal customers, lost customers and those at risk of churning.

The insights derived from RFM analysis can be used to guide personalized marketing strategies for improved sales and customer retention.

DATA OVERVIEW



The dataset contains three years of transactional data from the company, comprising 2747 rows and 20 columns, with each row representing a unique order. The key attributes include:

ORDERNUMBER: Unique identifier for each order.

QUANTITYORDERED: Number of items ordered in a specific transaction.

PRICEEACH: Price per unit of the product in the order.

ORDERLINENUMBER: Sequence number of the product in the order.

SALES: Total sales value for the order.

ORDERDATE: Date when the order was placed.

DAYS_SINCE_LASTORDER: Number of days since the customer's previous order.

STATUS: Current status of the order.

PRODUCTLINE: Product category to which the item belongs.

MSRP: Manufacturer's Suggested Retail Price for the product.

PRODUCTCODE: Unique identifier for the product.

CUSTOMERNAME: Name of the customer placing the order.

PHONE: Customer's contact phone number.

ADDRESSLINE1: Customer's primary address.

CITY: City of the customer's address.

POSTALCODE: Postal code of the customer's address.

COUNTRY: Country of the customer's address.

CONTACTLASTNAME: Last name of the customer's contact person.

CONTACTFIRSTNAME: First name of the customer's contact person.

DEALSIZE: Size category of the transaction.

EXPLORATORY DATA ANALYSIS

Summary Statistics

	count	unique		top	freq	mean	std	min	25%	50%	75%	max
QUANTITYORDERED	2747.0	NaN		NaN	NaN	35.103021	9.762135	6.0	27.0	35.0	43.0	97.0
PRICEEACH	2747.0	NaN		NaN	NaN	101.098951	42.042548	26.88	68.745	95.55	127.1	252.87
SALES	2747.0	NaN		NaN	NaN	3553.047583	1838.953901	482.13	2204.35	3184.8	4503.095	14082.8
DAYS_SINCE_LASTORDER	2747.0	NaN		NaN	NaN	1757.085912	819.280576	42.0	1077.0	1761.0	2436.5	3562.0
STATUS	2747	6		Shipped	2541	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PRODUCTLINE	2747	7		Classic Cars	949	NaN	NaN	NaN	NaN	NaN	NaN	NaN
PRODUCTCODE	2747	109		S18_3232	51	NaN	NaN	NaN	NaN	NaN	NaN	NaN
MSRP	2747.0	NaN		NaN	NaN	100.691664	40.114802	33.0	68.0	99.0	124.0	214.0
CUSTOMERNAME	2747	89	Euro Shopping Channel		259	NaN	NaN	NaN	NaN	NaN	NaN	NaN
CITY	2747	71		Madrid	304	NaN	NaN	NaN	NaN	NaN	NaN	NaN
COUNTRY	2747	19		USA	928	NaN	NaN	NaN	NaN	NaN	NaN	NaN
DEALSIZE	2747	3		Medium	1349	NaN	NaN	NaN	NaN	NaN	NaN	NaN

INFERENCES



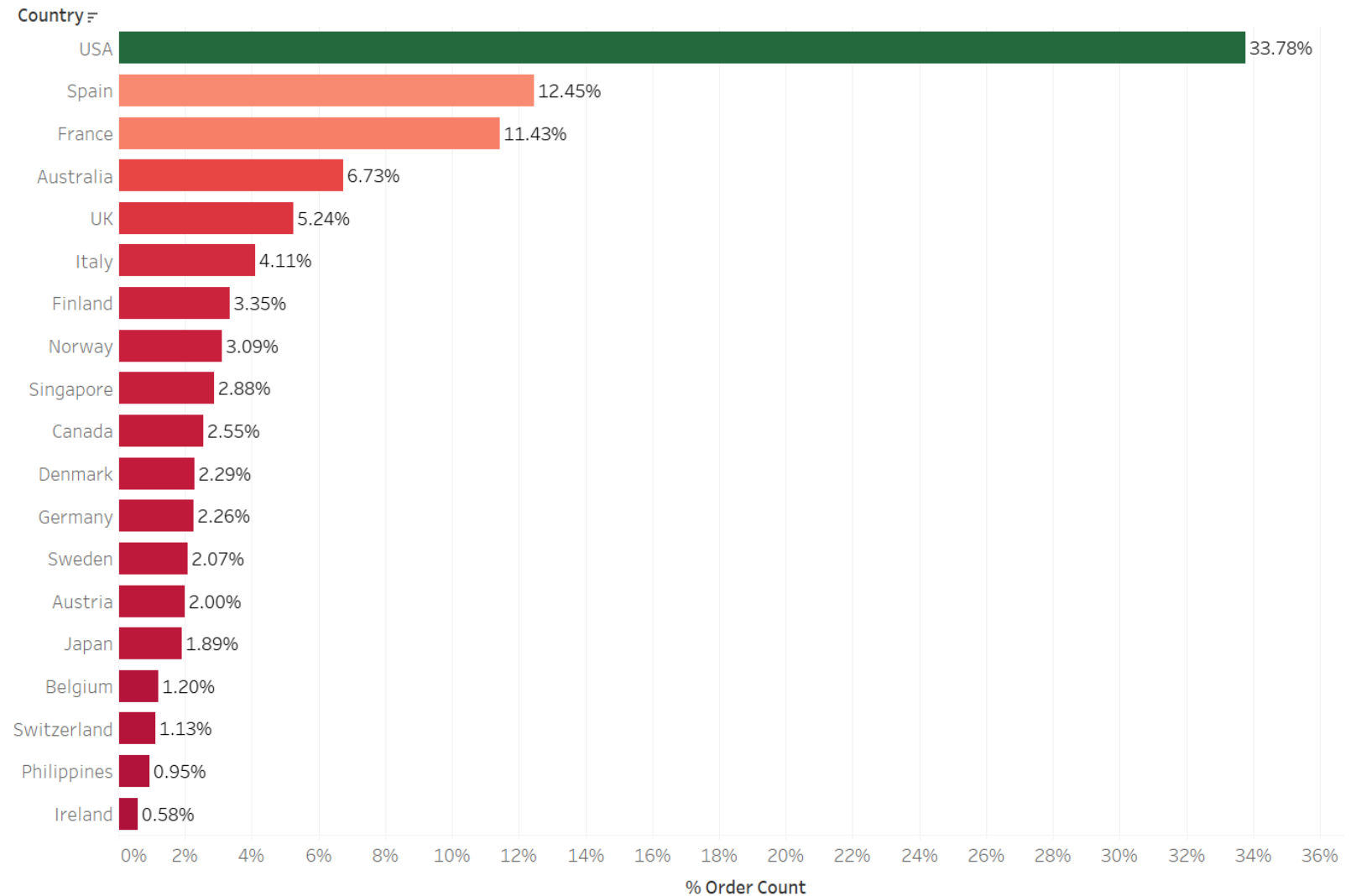
- **QUANTITYORDERED:** The quantity ordered ranges from 6 to 97, with 50% of the orders being for 35 units, indicating a common order size around this quantity.
- **PRODUCTLINE:** Classic Cars dominate the sales, with 949 units sold in this category being the highest.
- **PRODUCTCODE:** The product code S18_3232 is the most popular, with 51 sales recorded, suggesting it is highly favoured by customers.
- **PRICEEACH:** The price of automobile parts varies from 101 to 252, with 75% of parts priced at 127.1, indicating a typical price point around this value.
- **MSRP:** The manufacturer's suggested retail price ranges from 40 to 214, with an average of 99, showing a wide range in pricing with a moderate average.
- **DEALSIZE:** The majority of transactions are categorized as medium deals, totalling 1349.
- **CUSTOMERNAME:** The Euro Shopping Channel has the highest sales volume, with 259 transactions recorded.
- **COUNTRY:** The USA leads in sales, with 928 sales made, making it the largest market for the company.
- **CITY:** Madrid, Spain, ranks as the city with the highest sales, emphasizing the significance of the Spanish market.
- **SALES:** The sales of the automobile parts range from 1,838 to 14,082, with an average of 3,553, indicating variability in transaction sizes, but with a typical sales volume closer to the average.

UNIVARIATE ANALYSIS

1. Customer Details

Country:

The USA contributes to 33% of the customers, followed by Spain (12%), France (11%), Australia (6%), and UK (5%). Other countries account for less than 5% each, with Philippines and Ireland having very low contributions (<1%).

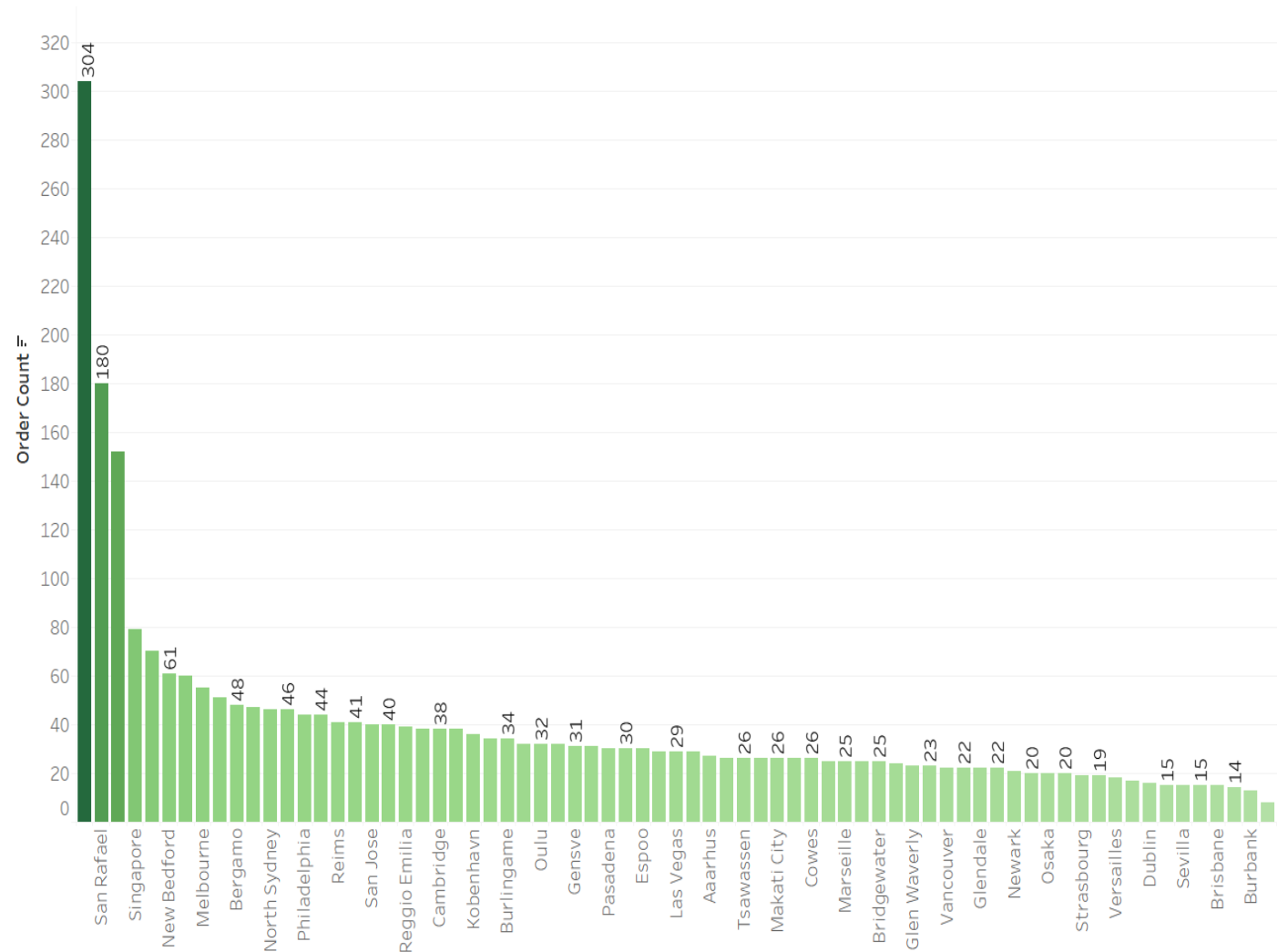


UNIVARIATE ANALYSIS

1. Customer Details

City:

Madrid is the top city with 304 transactions, followed by San Rafael (180) and NYC (152). Interestingly, Madrid is the only city in Spain with such high automobile parts purchases.



UNIVARIATE ANALYSIS

1. Customer Details

Customer Name:

Euro Shopping Channel is the highest customer with 259 transactions, followed by Mini Gifts Distributors with 180.

However, the transaction counts for other customers fall at 55 and below, which creates a significant distinction between these and the top customers.

Euro Shopping Channel 259	AV Stores, Co. 51	Saveley & Henriot, Co. 41	Technics Stores Inc. 34	Baane Mini Imports 32	Corrida Auto Replicas, Ltd 32	Oulu Toy Supplies, Inc. 32	Tokyo	Diecast Classics Inc. 31	Vida Sport, Ltd 31						
	Land of Toys Inc. 49	Salzburg Collectables 40	Toys of Finland, Co. 30		Amica Models & Co. 26	Cruz & Sons Co. 26					Mini Classics 26	Royal	Stylish Desk Decors, Co.		
	Muscle Machine Inc 48	The Sharp Gifts Warehouse 40			Toms Spezialitten, Ltd			Norway Gifts By Mail, Co.			Enaco	La Corne	Motor Mint		
	Rovelli Gifts 48	L'ordine Souvenirs 39												Herkku Gifts 29	
	Mini Gifts Distributors Ltd. 180	Anna's Decorations, Ltd 46	Scandinavian Gift Ideas 38	Signal Gift Stores 29		Collectable Mini Designs		Blauer See Auto, Co.					Lyon	Osaka	Gift Ideas Corp. 19
			Danish Wholesale Imports 36	UK Collectables, Ltd.				Canadian Gift							
		Souvenirs And Things Co. 46	Handji Gifts& Co 36	Auto Canal Petit		Marseille Mini Autos		Quebec Home		Mini Caravy 19					
Dragon Souvenirs, Ltd. 43		Mini Creations Ltd. 35	Heintze Collectables		Petit Auto 25		Classic Gift Ideas, Inc		Volvo Model						
Australian Collectors, Co. 55						Tekni Collectables		Auto Assoc. & Cie.				West Coast			
La Rochelle Gifts 53	Reims Collectables 41	Online Diecast Creations Co. 34	Marta's Replicas Co.	Vitachrome Inc.	Alpha Cognac		Super Scale Inc.								

UNIVARIATE ANALYSIS

2. Transaction Details

Order Date:

The highest number of transactions is seen in 2019, followed by 2018. Q4 shows consistent peak in transactions, while Q1 has the lowest. However, in 2020, Q1 experienced higher transaction volumes, although sales declined in Q2.

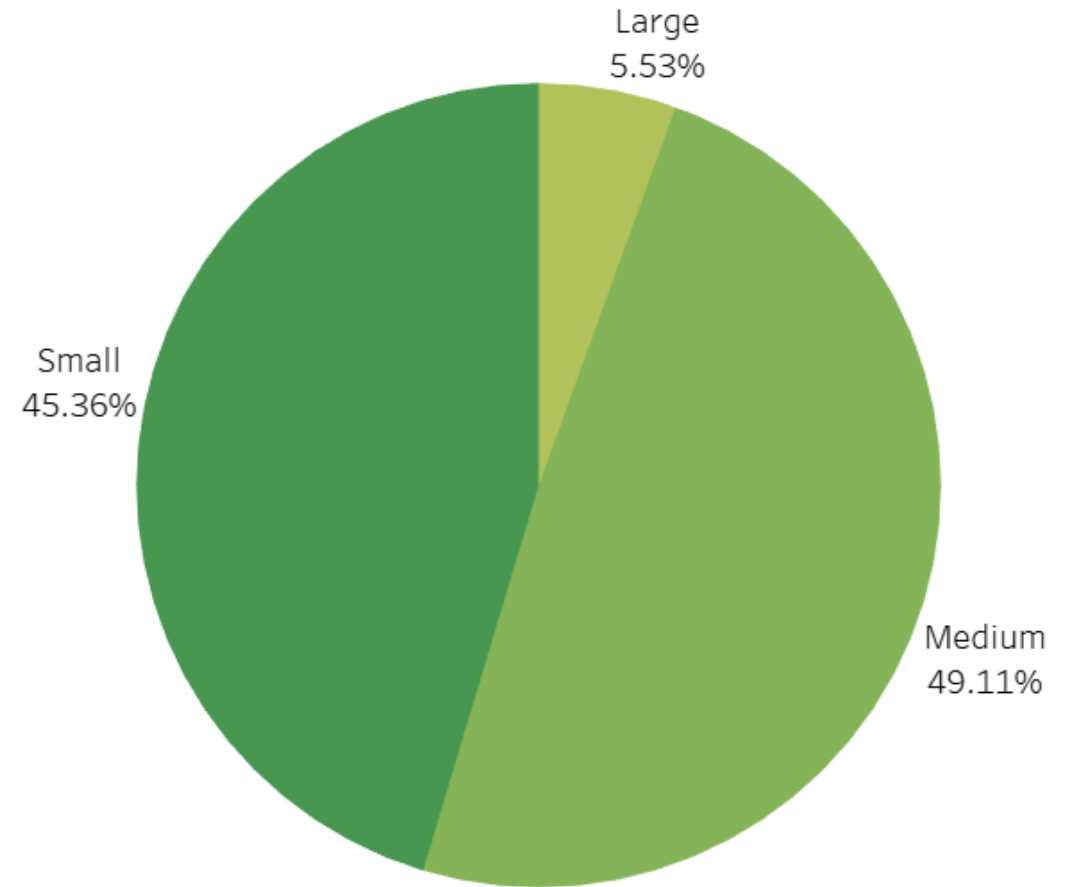


UNIVARIATE ANALYSIS

2. Transaction Details

Deal Size:

Most transactions are categorized as Medium (49%), followed by Small (45%) and Large deals (5%). Most customers place orders with moderate prices.

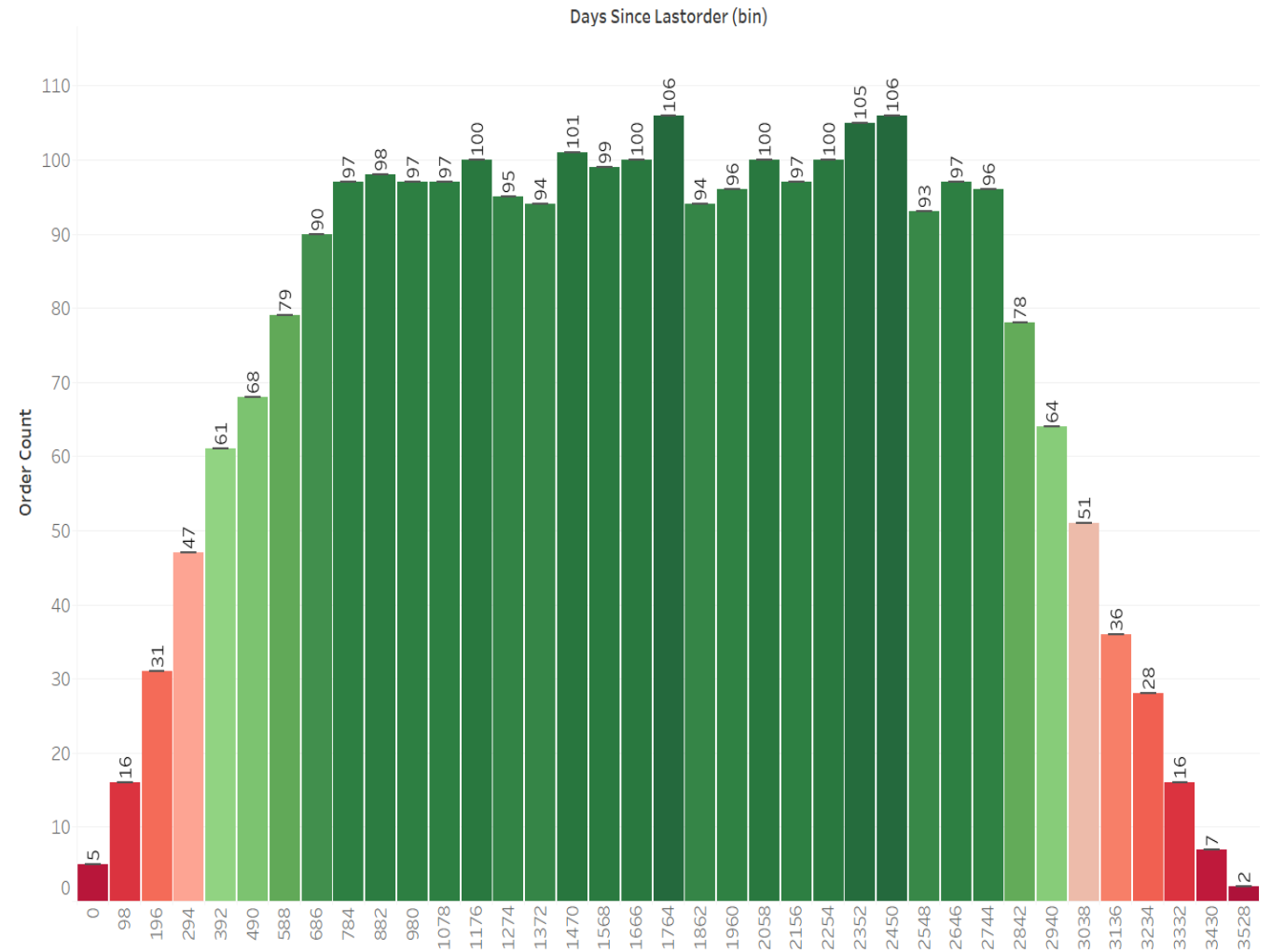


UNIVARIATE ANALYSIS

2. Transaction Details

Days Since Last Order:

The distribution of Days Since Last Order is symmetric, suggesting varying levels of recency among customers.



UNIVARIATE ANALYSIS

3. Product Information

Product Code:

The product code S18_3232 is the most popular, with 51 transactions, suggesting it is highly favoured by customers.

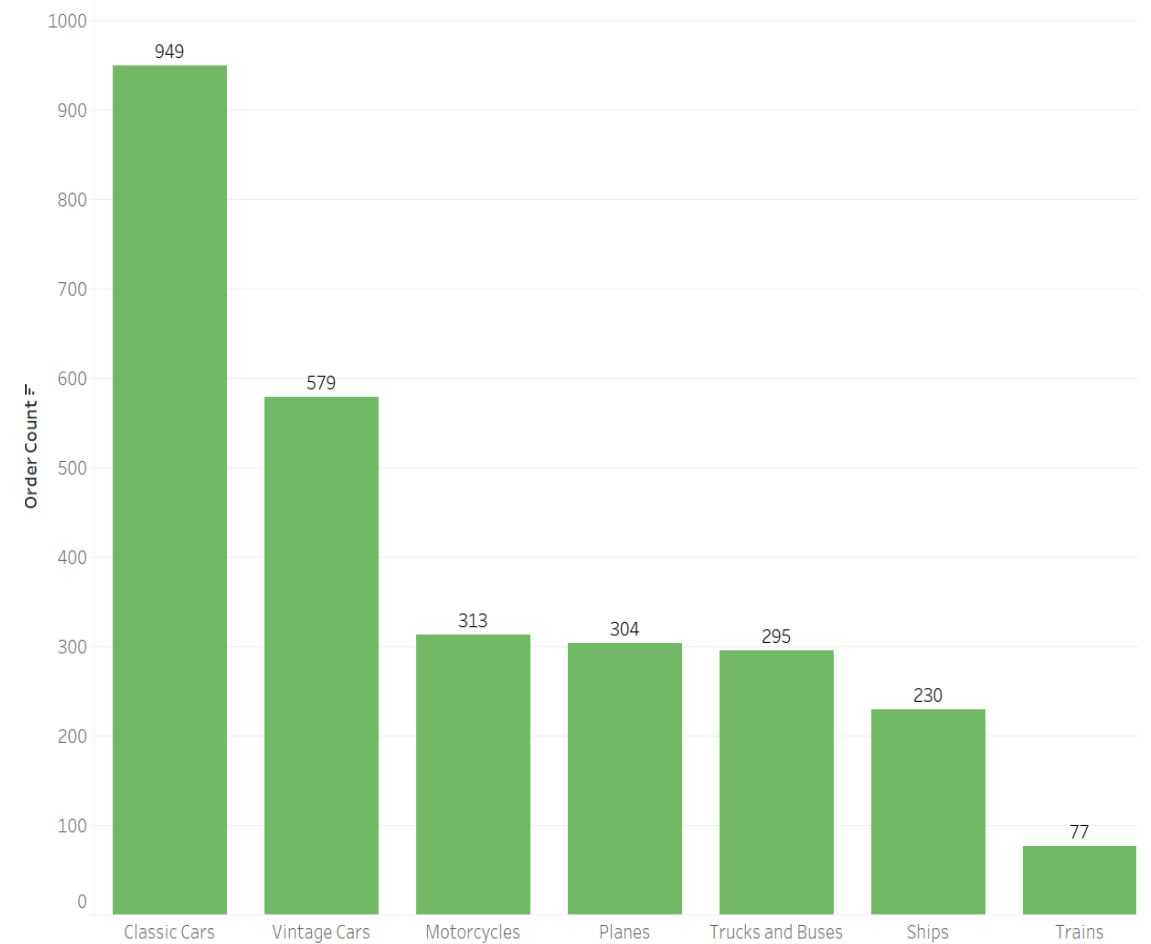
S18_3232 51	S18_4600 27	S12_3891 26	S24_4048 26											
	S24_2300 27	S18_1097 26	S32_3207 26											
S24_1444 28	S24_3949 27	S18_1662 26	S32_3522 26	S18_2949 25										
S24_2840 28				S18_2957 25										
S32_2509 28	S32_1268 27	S18_2319 26	S50_1341 26		S10_2016 24									
S50_1392 28	S700_2466 27	S18_4027 26	S50_1514 26	S18_3029 25										
S10_4962 27	S700_2824 27	S18_4522 26	S700_1138 26	S18_3136 25	S10_4698 24									
S12_4473 27				S18_3140 25	S12_1099 24	S24_2022 24								
S18_1129 27	S700_4002 27	S18_4668 26	S700_1691 26		S12_2823 24	S24_2360 24								
S18_1984 27	S10_1949 26	S24_2972 26	S700_2047 26	S18_3259 25	S12_3380 24	S24_4258 24	S18_2581 23							
S18_2238 27	S12_1108 26	S24_3420 26	S700_2610 26	S18_3685 25	S12_3990 24	S24_4278 24	S18_3278 23							
S18_2432 27	S12_1666 26	S24_3856 26	S700_3505 26	S18_3856 25	S18_1342 24	S32_1374 24	S18_4721 23	S24_4620 23						
				S24_2000 25	S18_2325 24	S32_2206 24	S24_1046 23	S18_4933 22						
							S24_1785 23	S24_2887 22						
									S18_4409					
										S24_3969				

UNIVARIATE ANALYSIS

3. Product Information

Product Line:

Classic Cars and Vintage Cars dominate sales, accounting for 34% and 21% respectively. Train parts have the lowest sales.

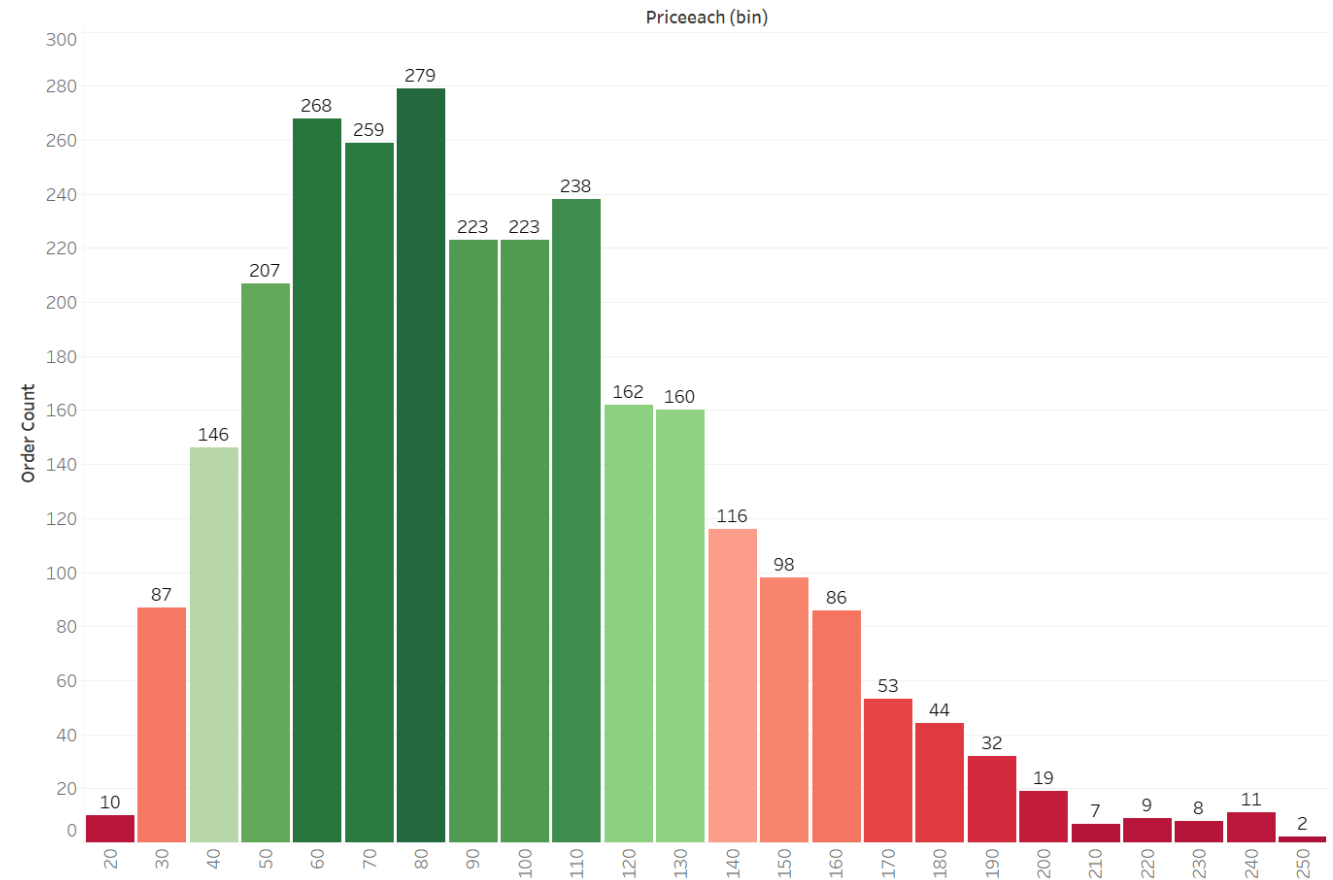


UNIVARIATE ANALYSIS

4. Pricing and Sales Data

Price Each:

Price Per Unit exhibit right-skewed distributions, with most values between 45-90 for MSRP and 50-110 for unit price. There are some outliers, which can be ignored as they represent extreme values.

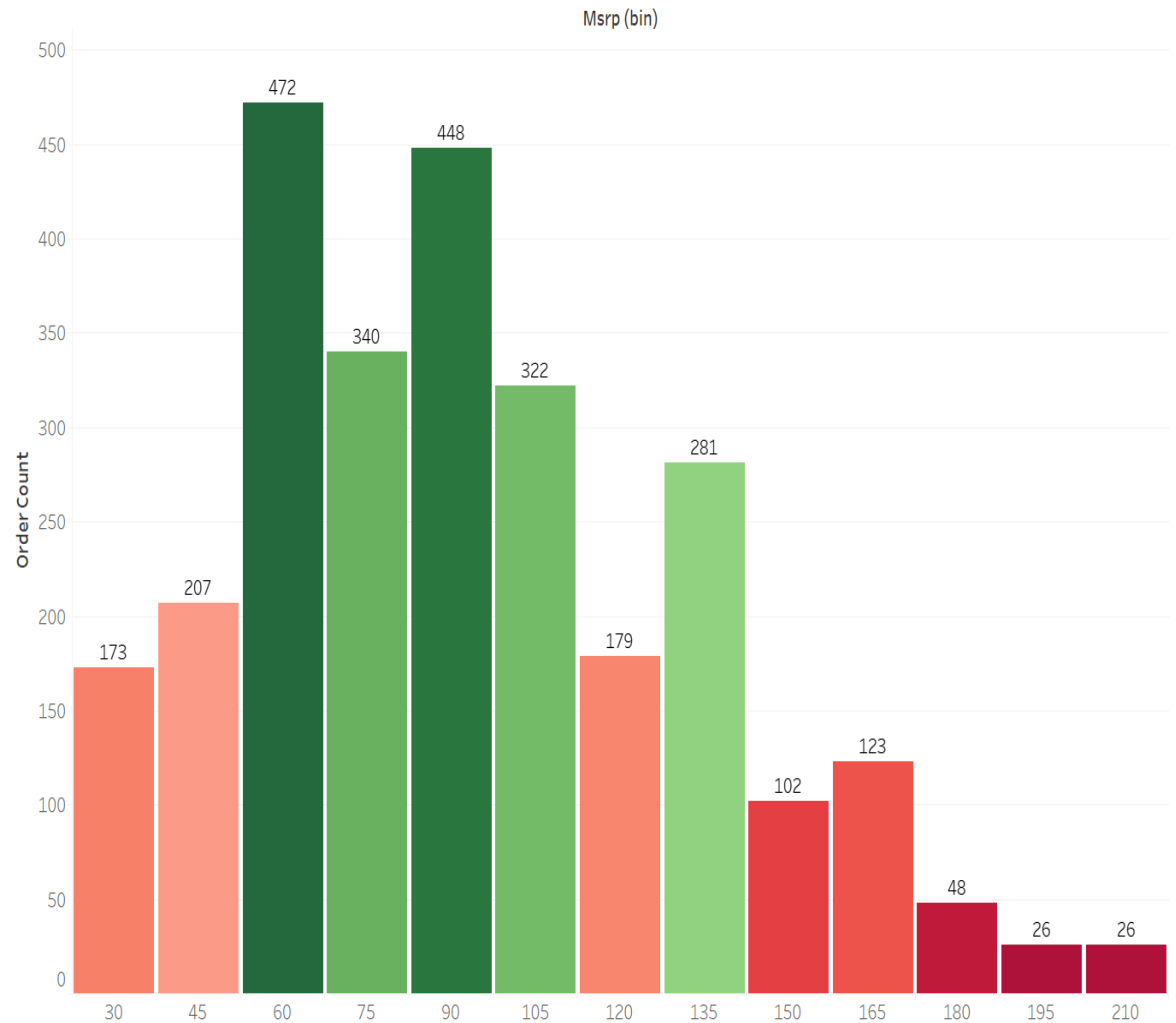


UNIVARIATE ANALYSIS

4. Pricing and Sales Data

MSRP:

Price Per Unit exhibit right-skewed distributions, with most values between 45-90 for MSRP and 50-110 for unit price. There are some outliers, which can be ignored as they represent extreme values.

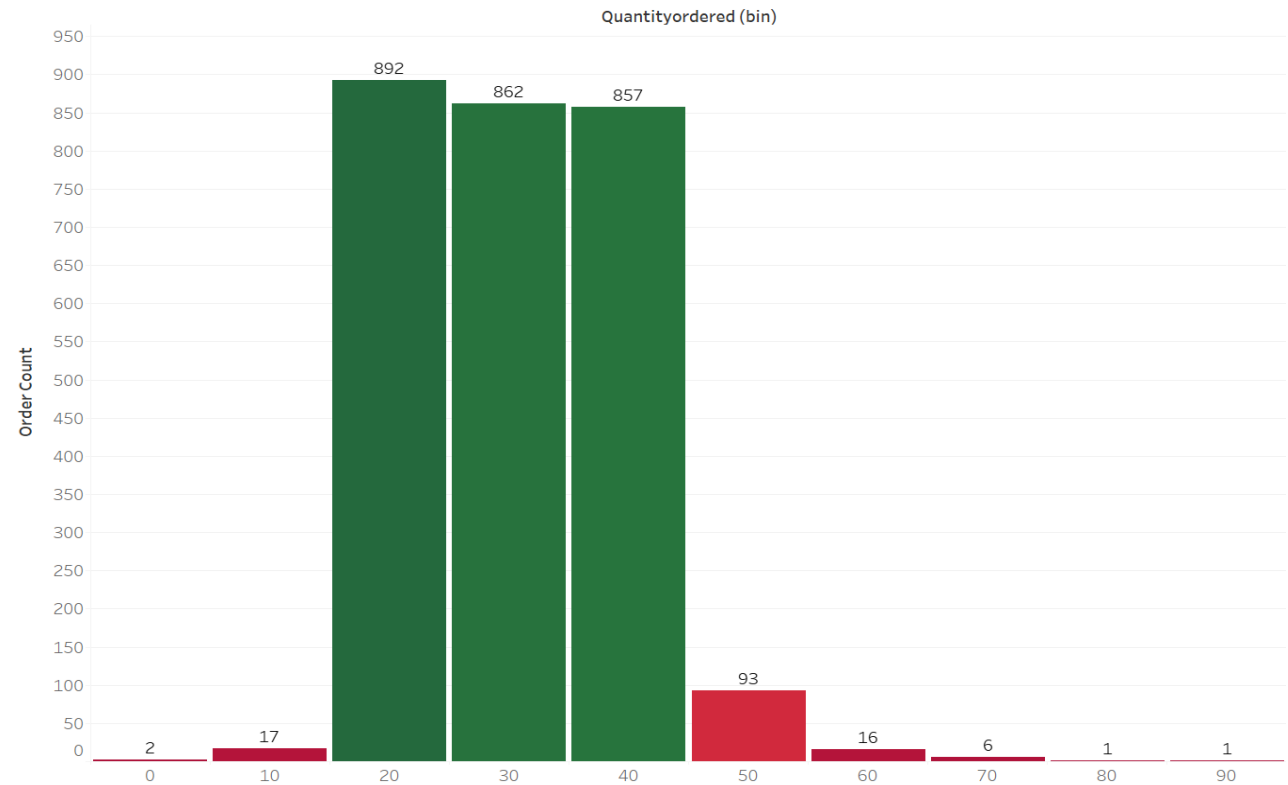


UNIVARIATE ANALYSIS

4. Pricing and Sales Data

Quantity Ordered:

The most common order quantities are between 20-40 (1754 transactions) and 40-60 (950 transactions), suggesting that the majority of customers tend to place medium-sized orders.

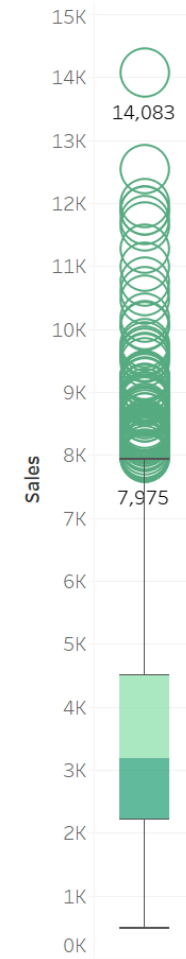


UNIVARIATE ANALYSIS

4. Pricing and Sales Data

Sales:

Sales range from 400-8000, with 50% of sales falling between 2000-4500. This indicates that most transactions are of medium-sized deals.



BIVARIATE ANALYSIS

1. Sales Correlations

Sales by Quantity Ordered:

Sales range from 400-8000, with 50% of sales falling between 2000-4500. This indicates that most transactions are of medium-sized deals.

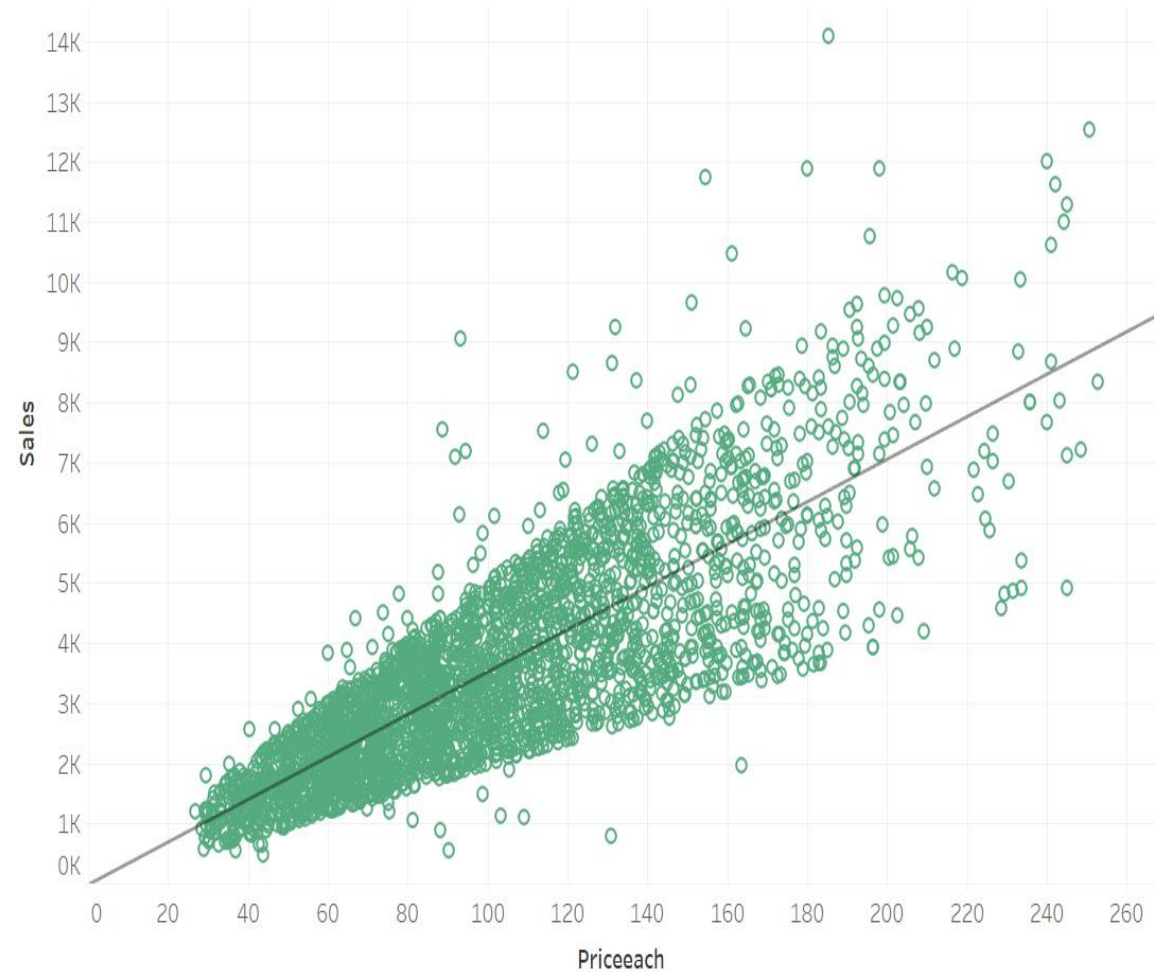


BIVARIATE ANALYSIS

1. Sales Correlations

Sales by Price each:

Sales show a positive correlation with Quantity Ordered and Price each meaning that higher quantities and prices generally lead to higher sales.



BIVARIATE ANALYSIS

1. Sales Correlations

Sales by MSRP:

Sales show a positive correlation with MSRP which suggests that high-priced products are doing better than expected.

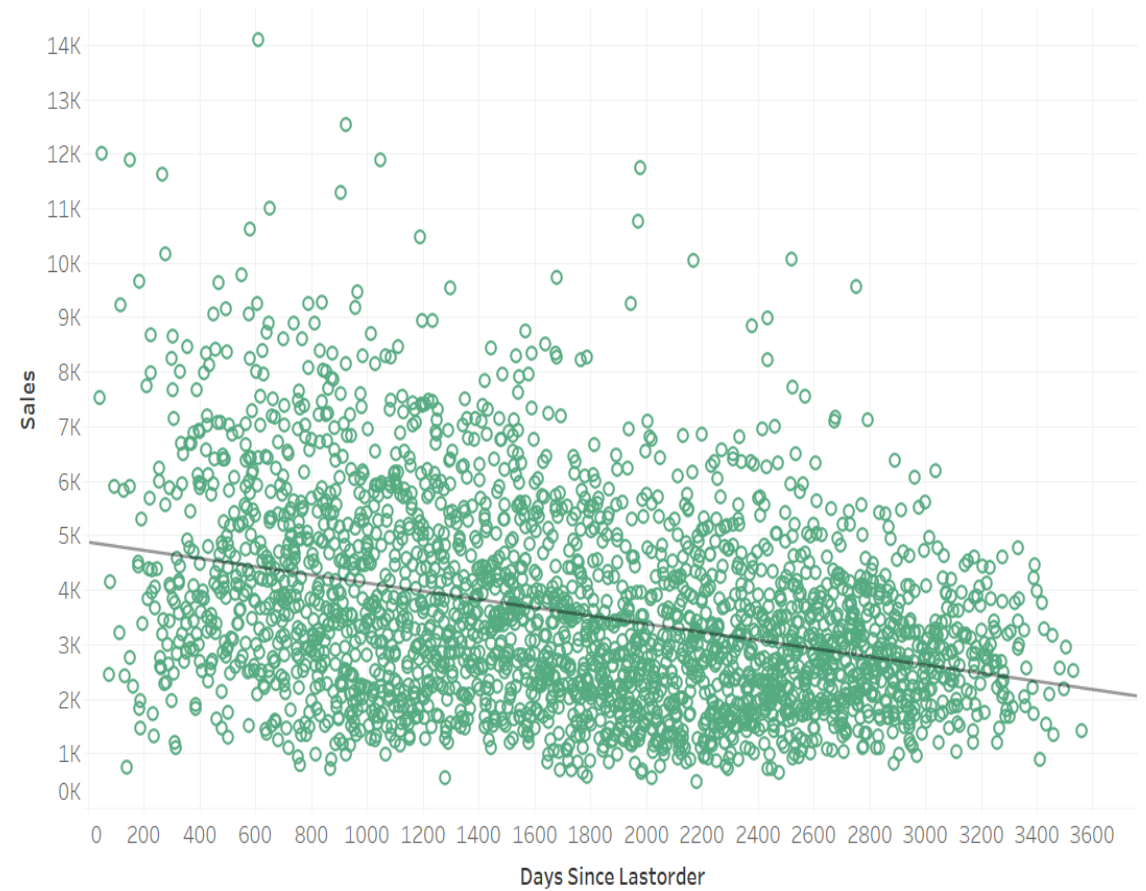


BIVARIATE ANALYSIS

1. Sales Correlations

Sales by Days since last order:

Sales exhibit a negative trend with Days Since Last Order, suggesting that the more recent the order, the higher the sales.

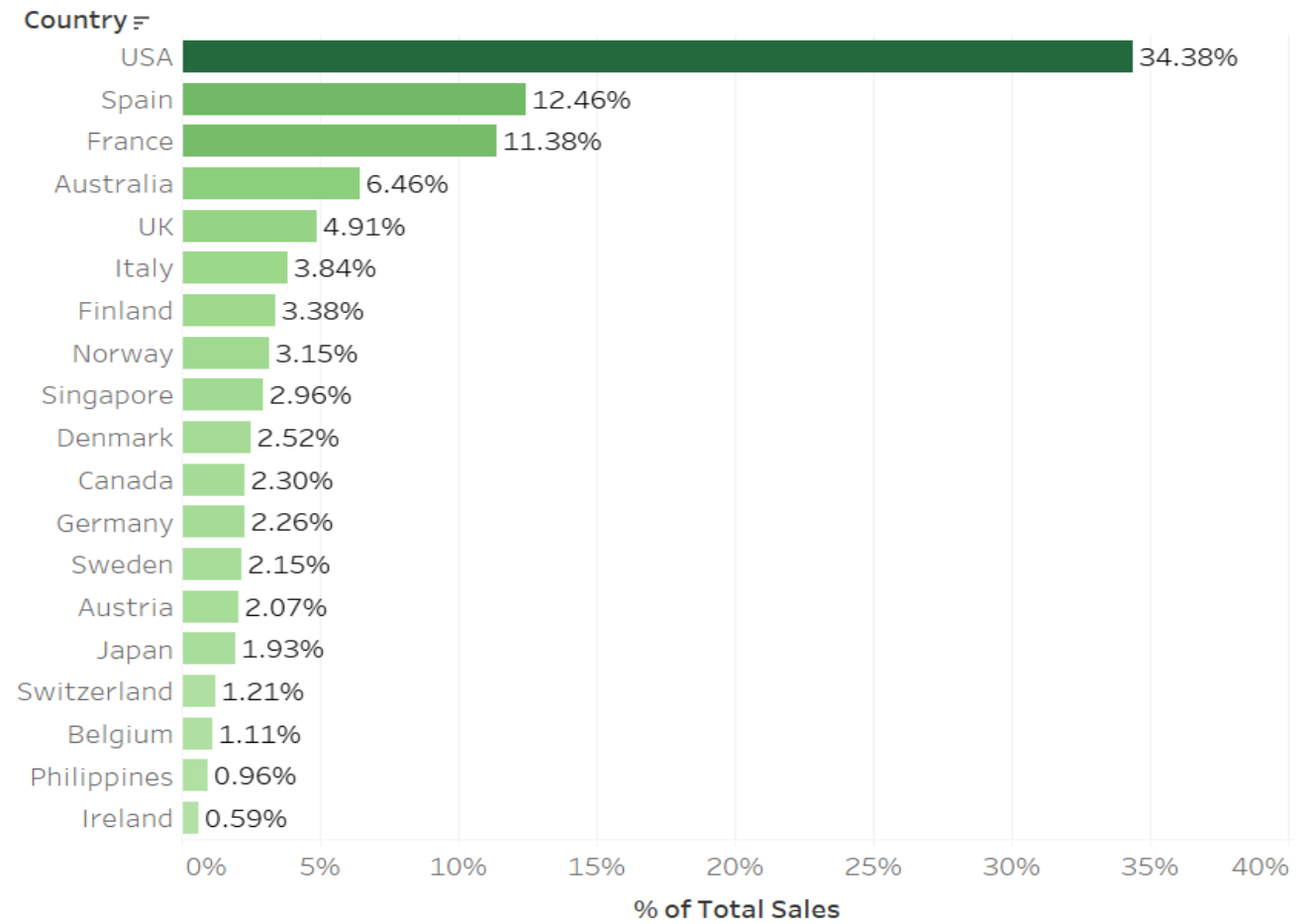


BIVARIATE ANALYSIS

2. Geographic Insights

Sales by Country:

USA, Spain, France, Australia, and UK contribute significantly to sales, with USA leading.

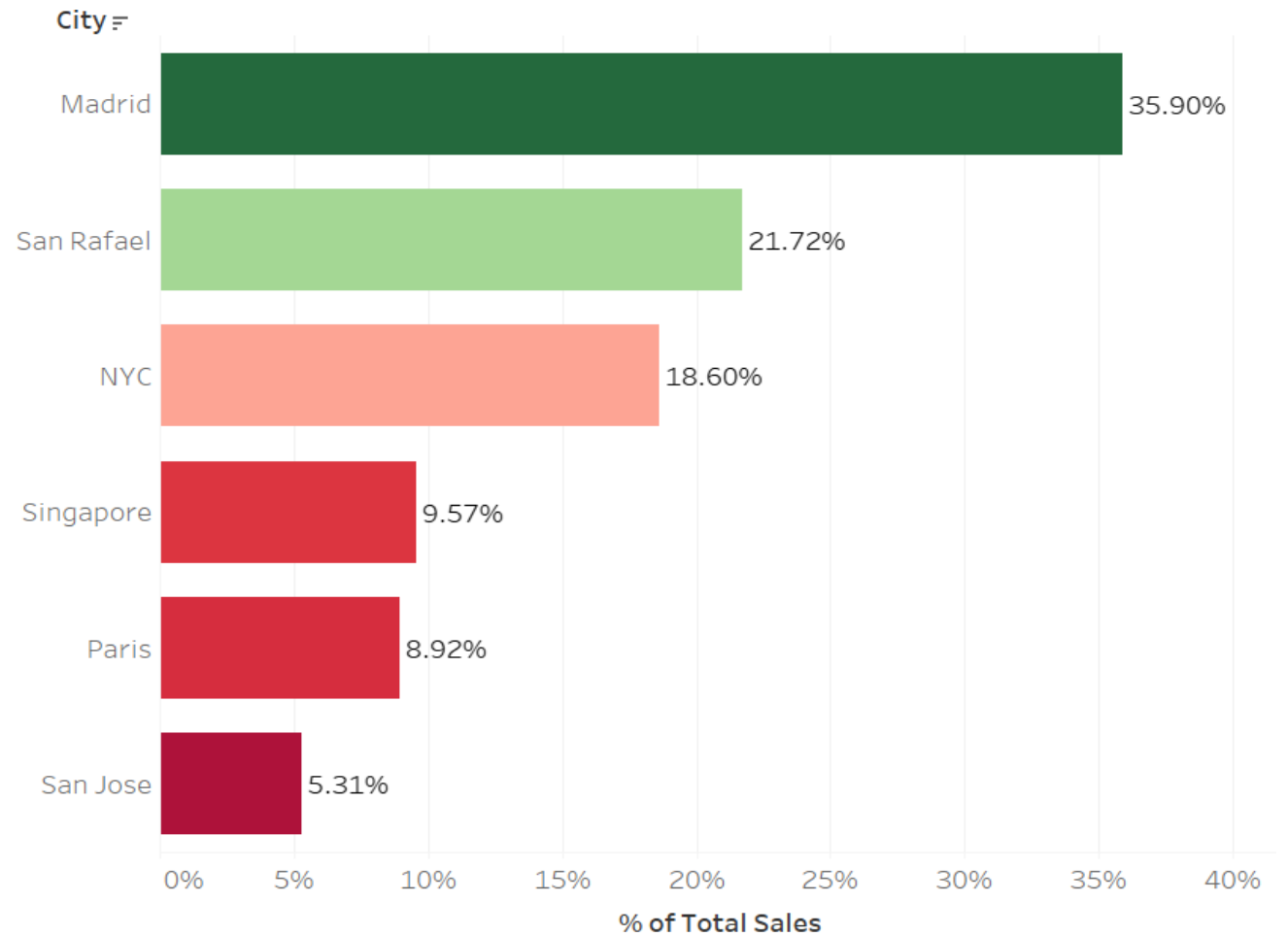


BIVARIATE ANALYSIS

2. Geographic Insights

Sales by City:

Small and Medium Deals are evenly distributed across cities, while Large Deals are primarily concentrated in the USA, France, and Spain. Madrid stands out for high sales across all deal sizes, and large deals could be extended to cities like San Rafael, NYC, Paris and San Jose.

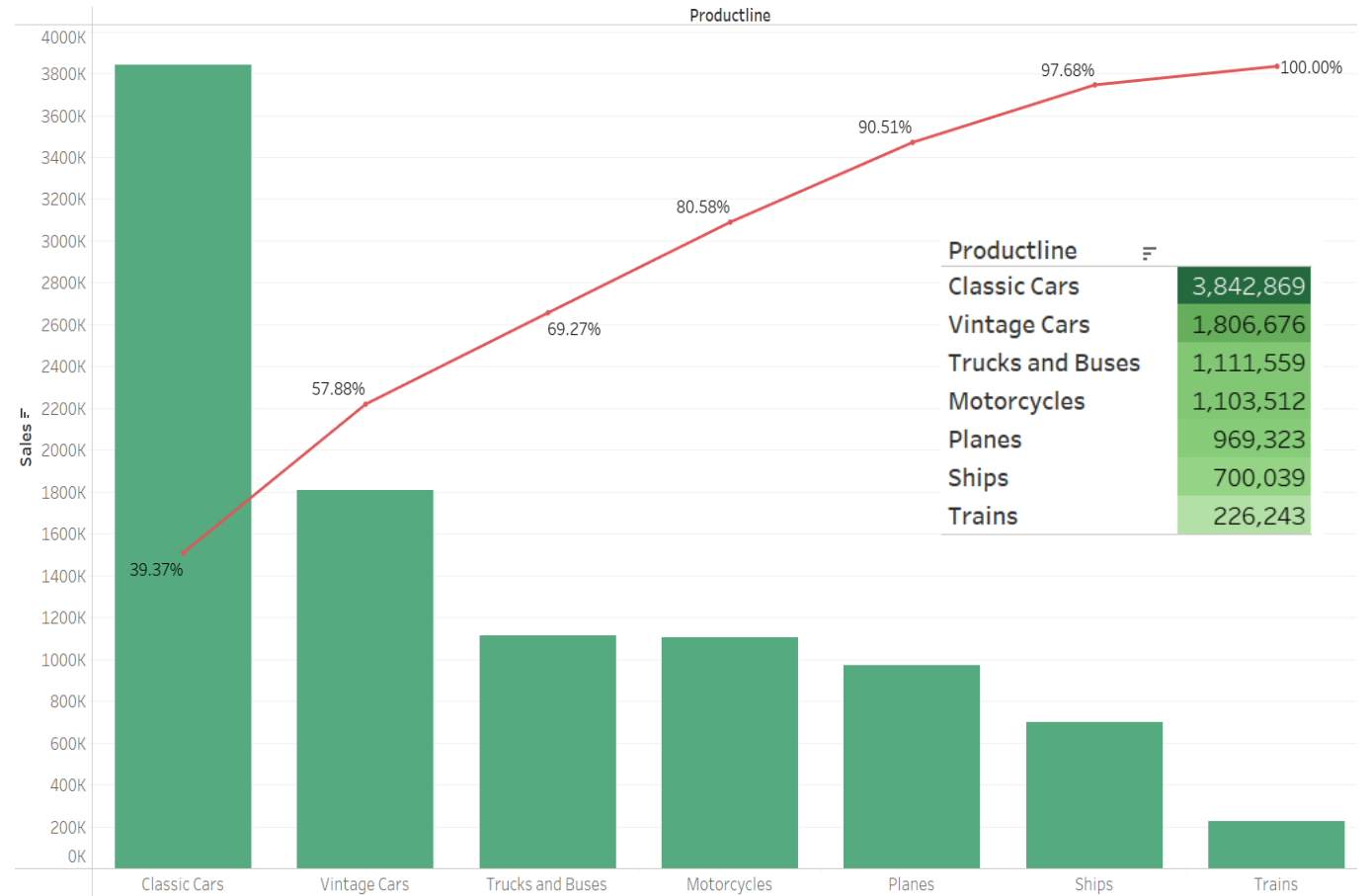


BIVARIATE ANALYSIS

3. Product Categories

Sales by Product Line:

Classic Cars and Vintage Cars are the primary contributors to sales, while Buses and Trucks and Motorcycles also play important supporting roles in generating significant revenue.

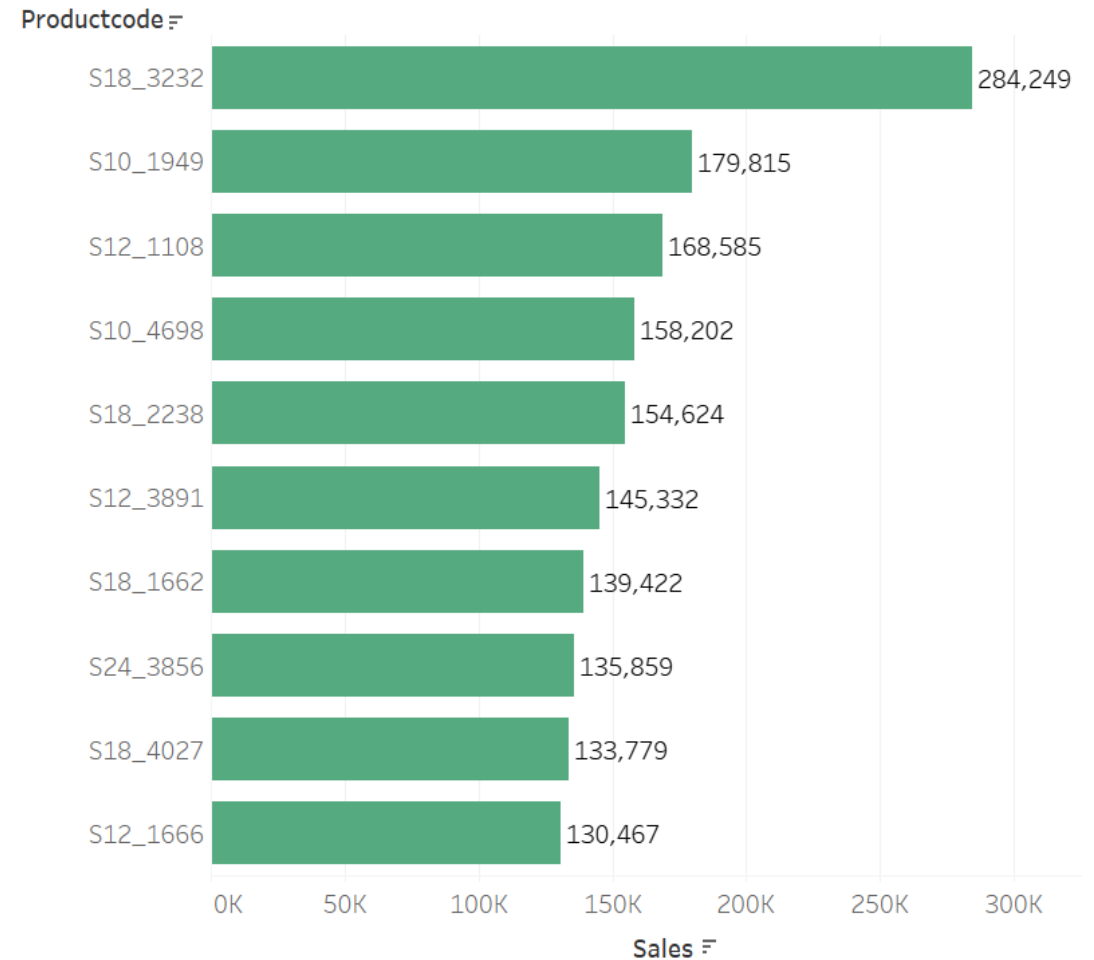


BIVARIATE ANALYSIS

3. Product Categories

Sales by Product Code:

Classic Cars with product codes S18_3232, S10_1949, and S12_1108 are the top sellers.

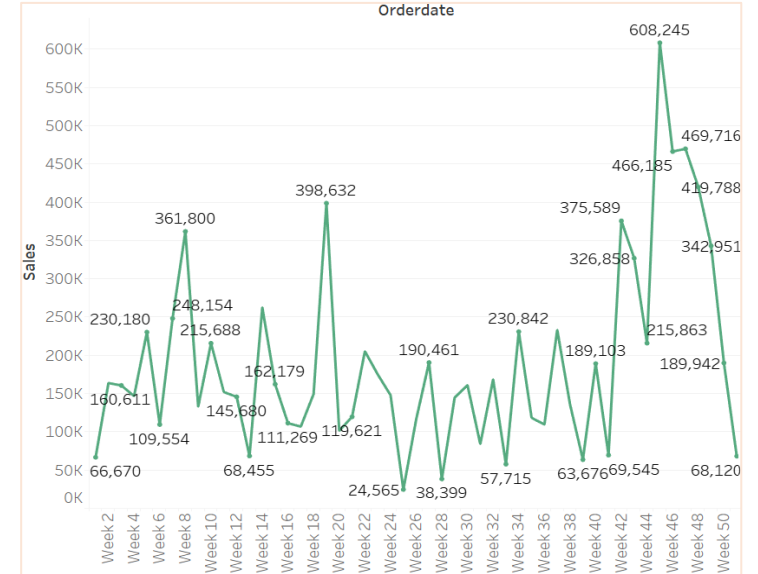
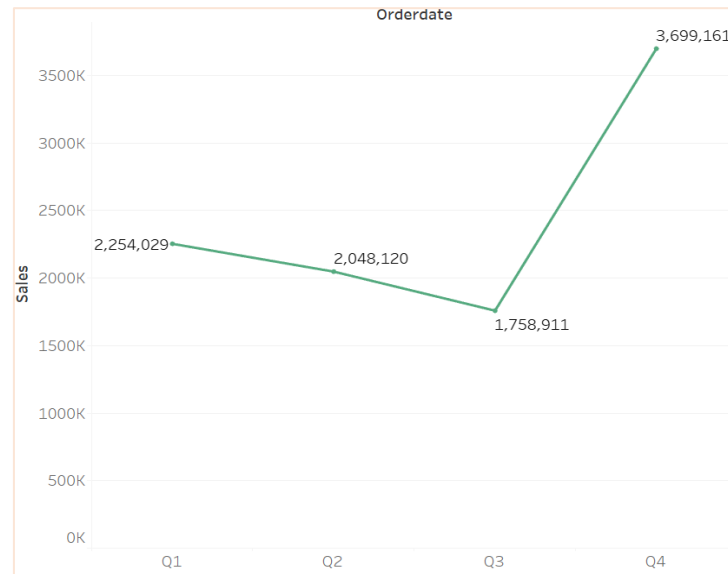
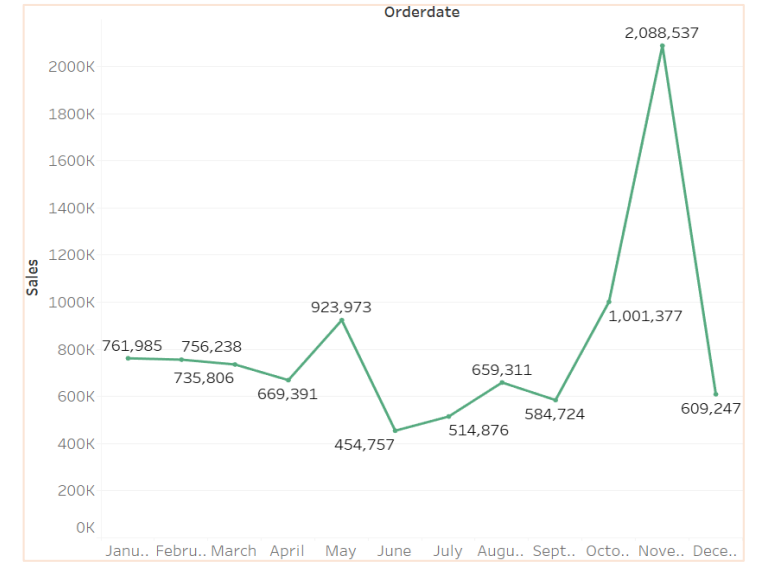
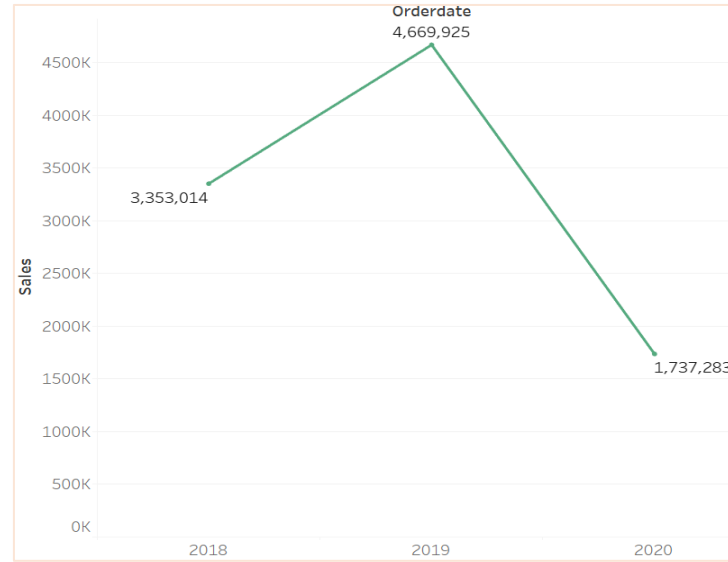


BIVARIATE ANALYSIS

4. Sales Trends over Time

2019 had the highest sales, while 2020 experienced the lowest sales, consistent across deal sizes and product lines.

Sales peak during weeks 42-43 and 45-49, and October-November shows the highest sales across all deal sizes. Q4 sees the highest sales, while Q3 has the lowest.



BIVARIATE ANALYSIS

4. Price Sensitivity Analysis

The quantity ordered shows a constant trend with both MSRP and Price Each, which suggests that the quantity of products ordered remains relatively stable as both the MSRP and the price each change.

Customers may have a fixed buying pattern where they consistently order a set quantity of products, regardless of the price. This can occur if the product is used regularly or in bulk, so the order size remains the same.

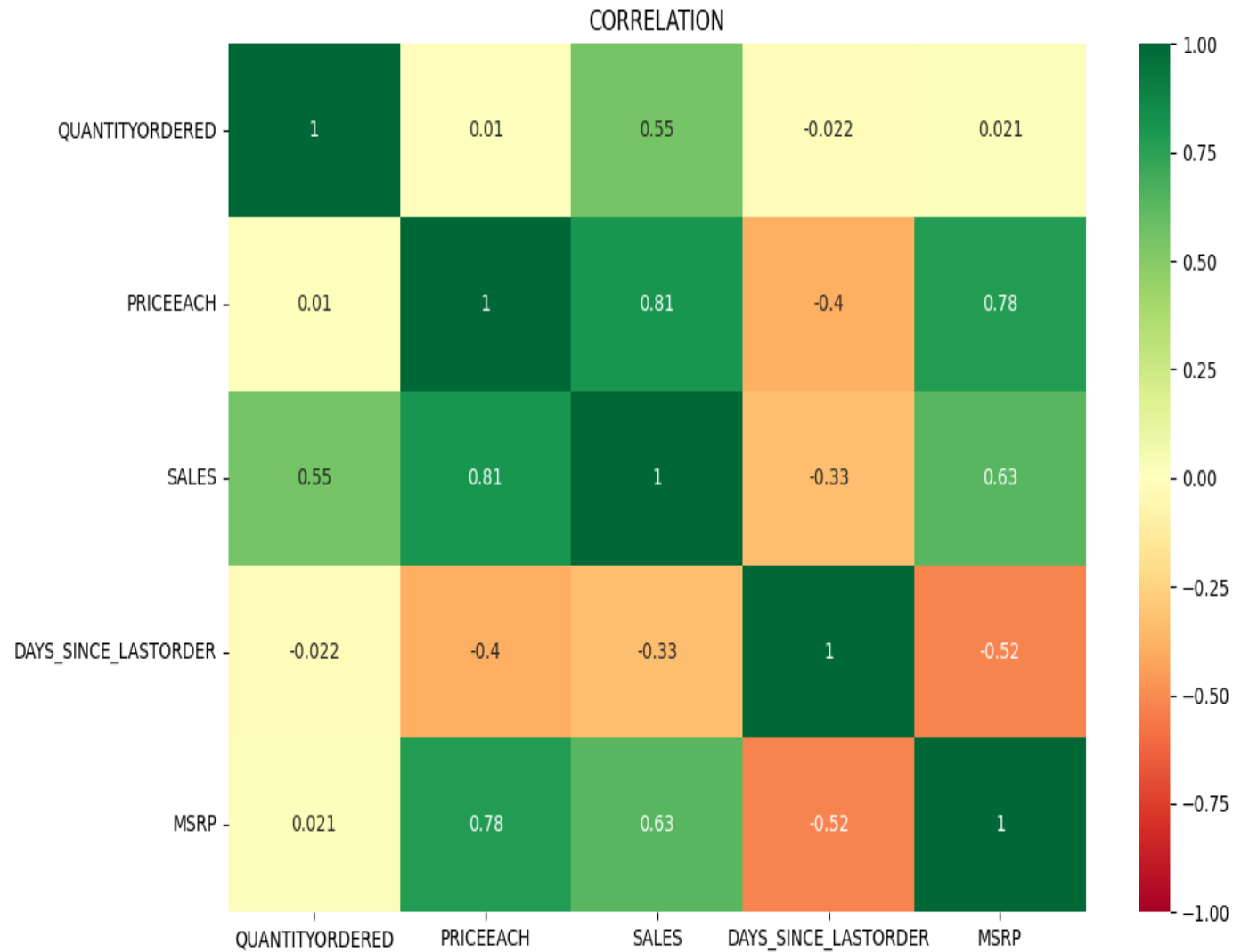


MULTIVARIATE ANALYSIS

1. Correlation between Numeric Variables

Positive correlations (Sales & Price, Sales & MSRP, Sales & Quantity Ordered) indicate that higher-value products or those ordered in bulk are typically associated with increased sales.

Negative correlations (MSRP & Days Since Last Order, Sales & Days Since Last Order) suggest that items with higher prices or lower sales tend to have longer gaps between orders, which could reflect less frequent customer interest or purchasing behaviour.



MULTIVARIATE ANALYSIS

2. Top Customers by Sales and Quantity ordered

The top customers driving the highest sales and order volume are Euro Shopping Channel (Spain), Mini Gift Distributors (USA), Australian Collectors (Australia), La Rochelle Gifts (France), and Muscle Machine (USA). Additionally, higher sales are associated with larger quantities ordered.

While Singapore, Denmark, Italy, and Austria may not contribute significantly to overall sales, they still have customers who rank among the top performers in terms of sales.

Euro Shopping Channel 912,294 9,327 Spain	Australian Collectors, Co. 200,995 1,926 Australia	La Rochelle Gifts 180,125 1,832 France	AV Stores, Co. 157,808 1,778 UK	Muscle Machine Inc 197,737 1,775 USA
	The Sharp Gifts Warehouse 160,010 1,656 USA	Dragon Souvenirs, Ltd. 172,990 1,524 Singapore	Anna’s Decorations, Ltd 153,996 1,469 Australia	Salzburg Collectables 149,799 1,442 Austria
	Land of Toys Inc. 164,069 1,631 USA			
Mini Gifts Distributors Ltd. 654,858 6,366 USA	Souvenirs And Things Co. 151,571 1,601 Australia	Saveley & Henriot, Co. 142,874 1,428 France		L’ordine Souvenirs 142,601 1,280 Italy
		Danish Wholesale Imports 145,042 1,315 Denmark		

MULTIVARIATE ANALYSIS

3. Sales Distribution by Order Line Number

There is a decreasing trend in sales as the order line number increases, which indicates a preference for certain products, with earlier products in the order generating more sales.

The customers are more selective or focused on purchasing the higher-priority or high-demand items at the beginning of the order, while adding less popular or less essential items later.

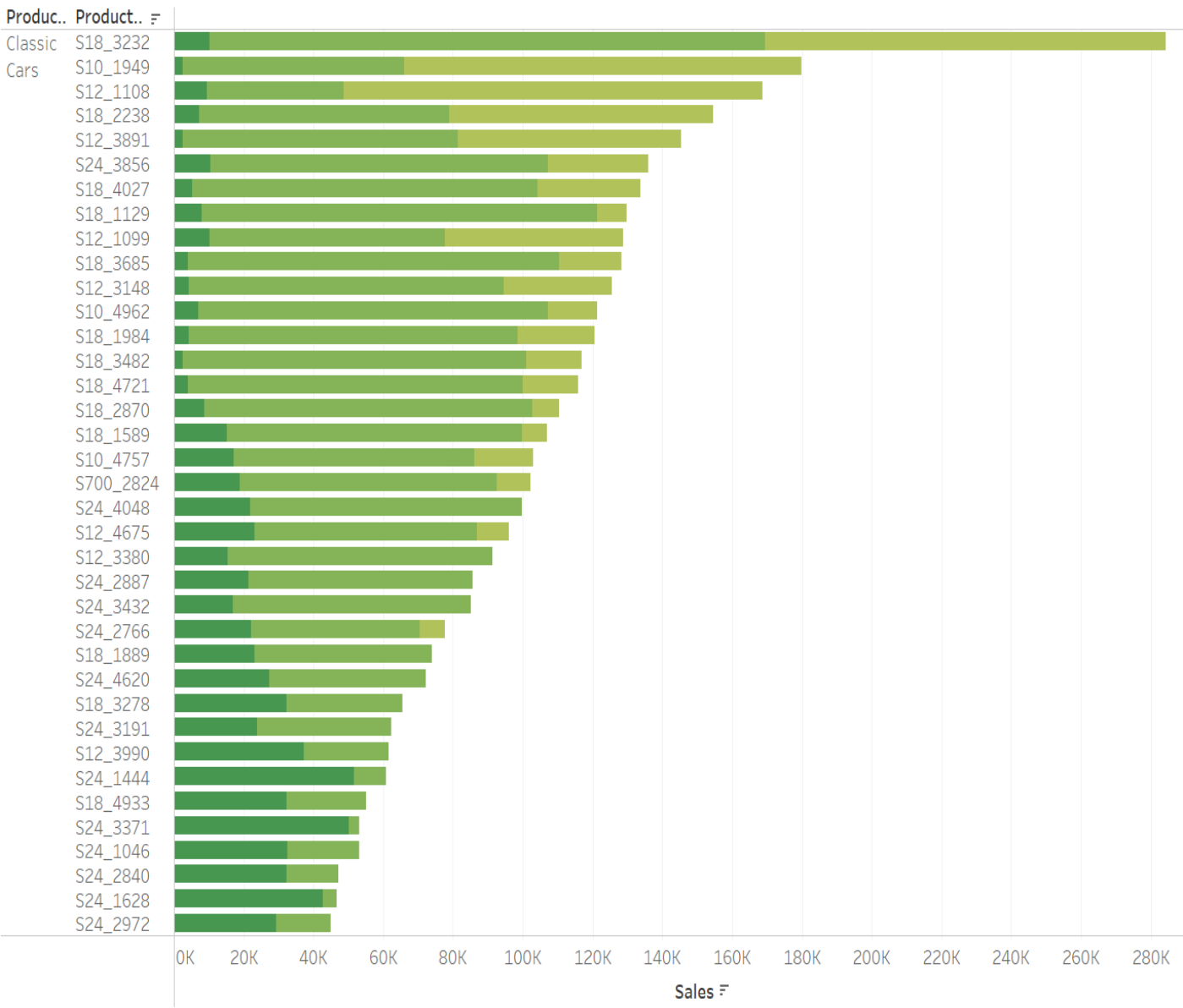
Productline	Orderlinenumber																	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Classic Cars	3,564	3,452	3,052	3,097	2,721	2,971	1,927	2,276	2,044	1,812	1,749	1,331	1,150	888	744	407	188	
Vintage Cars	2,329	2,464	1,807	1,505	1,605	1,770	1,482	1,242	1,142	726	896	818	678	728	229	337	261	40
Motorcycles	1,512	1,342	1,193	1,170	1,194	910	629	626	531	325	322	408	297	242	145	137	97	
Planes	1,017	912	931	938	686	565	566	852	686	696	569	535	343	417	381	275	178	89
Trucks and Buses	1,053	835	1,205	794	1,224	656	888	993	587	829	476	356	318	209	46	47	32	31
Ships	806	762	609	806	767	547	721	382	475	469	245	214	433	142	229	198	31	153
Trains	215	222	170	472	86	255	251	92	155	60	217	26	69	159	131	42	90	

MULTIVARIATE ANALYSIS

4. Sales Breakdown Across Product Code, Product Line, and Deal Size-Classic Cars

Classic Cars lead the sales, accounting for 40% of total sales, with top-selling products like S18_3232, S10_1949, and S12_1108 driving much of this success. These products not only generate high sales but also show a significant concentration of medium and large transactions. This suggests that customers are more likely to make higher-value purchases when selecting these products.

Given their strong performance in terms of both sales and transaction size, these top-selling Classic Cars could be strategic choices for targeting large-value transactions.

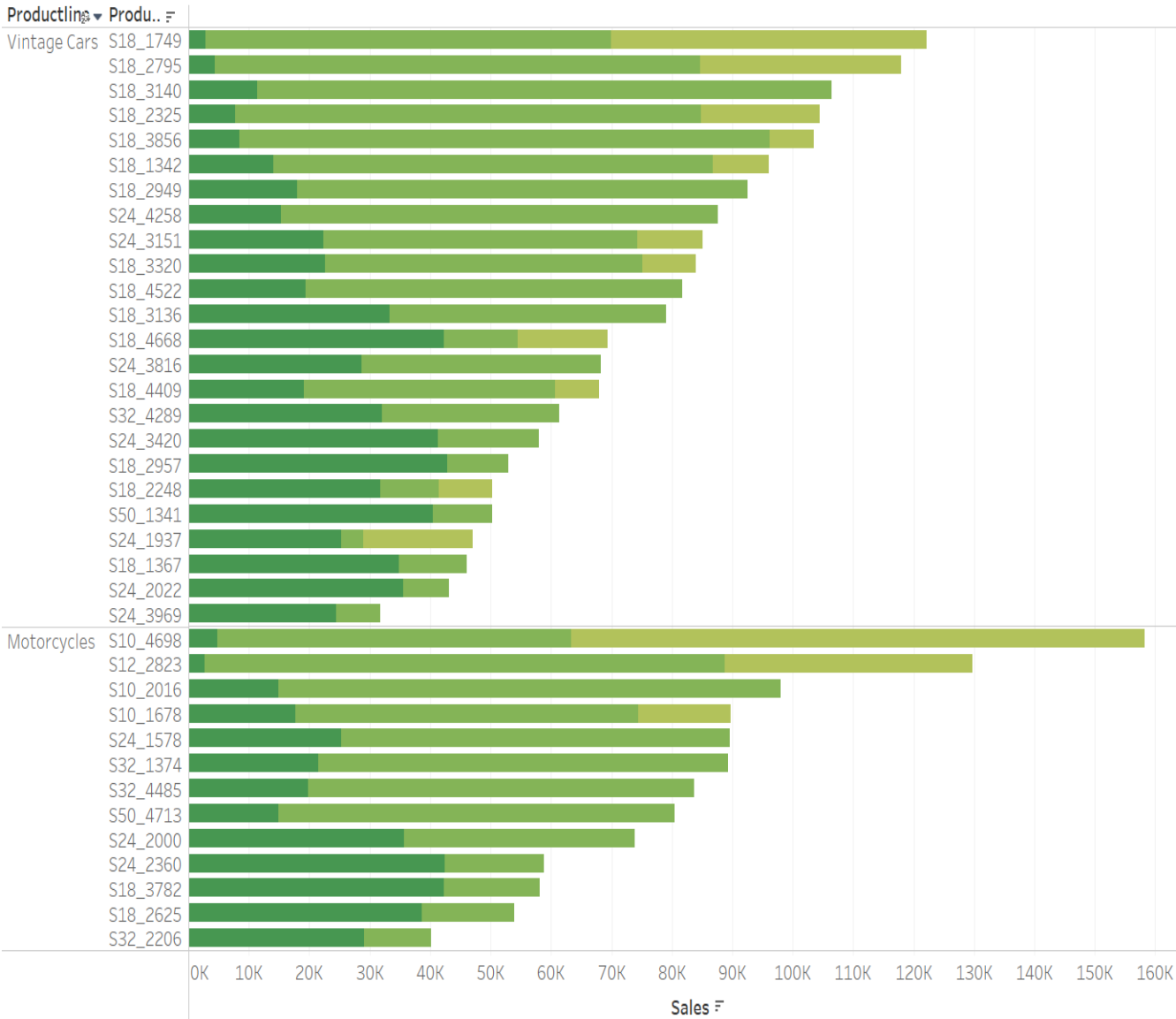


MULTIVARIATE ANALYSIS

5. Sales Breakdown Across Product Code, Product Line, and Deal Size- Vintage Cars and Motorcycles

Following Classic Cars, Vintage Cars (such as S18_1749 and S18_2795) and Motorcycles (including S10_4698 and S12_2823) also emerge as key contributors to overall sales. Notably, S10_4698 stands out as a strong performer, particularly for large transactions. This product has proven to be a preferred choice for customers looking to make substantial purchases, further reinforcing its value in driving high revenue.

The correlation between sales volume and transaction size is evident, as products with higher sales tend to be linked with larger transactions, significantly impacting the overall revenue.



OBSERVATIONS AND INSIGHTS FROM EDA

Geographic Distribution:

- The USA emerges as the dominant region, accounting for 33% of the customer base. This is significant, indicating that a substantial portion of the business is driven by American customers, and therefore, focusing marketing and sales efforts in this region would be crucial.
- Spain and France follow, with 12% and 11% of the customers, respectively. Madrid stands out as the only city in Spain with notably high sales of automobile parts, which could be an indication of a regional hub or specialized demand in this city.
- Countries like the Philippines and Ireland contribute very little (<1%), suggesting that marketing efforts in these regions may be less effective or underdeveloped, presenting opportunities for targeted growth.

Customer Behavior:

- The distribution of deal sizes shows that Medium deals dominate the market, representing 49% of the total transactions. This indicates that the typical transaction falls into a middle-range price point, which can be important for businesses to focus on as they likely generate consistent revenue.
- Small deals make up 45%, with Large deals contributing only 5%. Although most transactions are on the smaller side, focusing on increasing the share of large transactions could significantly boost overall revenue, especially considering the high-value products like Classic Cars.
- Euro Shopping Channel (Spain) is the leading customer, with the highest number of transactions (259). This reinforces the importance of nurturing strong relationships with top-performing customers who drive a large portion of the sales.

Product Preferences:

- Classic Cars and Vintage Cars dominate the sales mix, contributing 40% and 21% of total sales, respectively. The products S18_3232, S10_1949, and S12_1108 represent the top-sellers within the Classic Cars category. Given that these products generate both high sales and larger transaction sizes, they are ideal candidates for focusing on high-value marketing and sales strategies.
- Motorcycles also contribute significantly, with products like S10_4698 showing strong performance in large transactions. This suggests that motorcycles, while not leading in total sales, attract customers making substantial purchases.

OBSERVATIONS AND INSIGHTS FROM EDA

Sales Trends:

- Sales are highest in Q4, particularly during weeks 42-43 and 45-49, and October-November shows the peak. This could indicate seasonal demand spikes or specific marketing activities that influence purchasing behavior during this time. I
- In contrast, 2020 saw a significant drop in sales compared to 2019, which had the highest sales. This decrease could be due to external affecting global purchasing patterns. Understanding these fluctuations is important for businesses to adjust their sales forecasts and strategies based on market conditions.

Correlation Insights:

- Positive correlations between Sales, Quantity Ordered, MSRP, and Price Per Unit suggest that customers who buy in larger quantities or higher-priced products tend to generate higher sales. This reinforces the idea that premium products (like Classic Cars) and bulk purchases drive substantial revenue.

Customer Segmentation:

- Top customers such as Euro Shopping Channel (Spain) and Mini Gift Distributors (USA) are the biggest drivers of sales and order volume. This makes them crucial for targeting account-based marketing efforts, special offers, or loyalty programs to strengthen the relationship and increase lifetime value.
- Smaller regions like Singapore, Denmark, Italy, and Austria may not contribute significantly to overall sales, but they still have high-value customers within their ranks. This could indicate untapped potential for growth in these areas or niches that require focused outreach.

OBSERVATIONS AND INSIGHTS FROM EDA

Order Behavior:

- There's a decreasing trend in sales as the order line number increases. This could indicate that customers tend to prioritize purchasing high-demand products (such as Classic Cars) at the beginning of their order, and less popular items are ordered later.
- This behaviour highlights the opportunity to optimize product placements in order forms or suggest high-value items first to increase the likelihood of larger sales.

Pricing Trends:

- Both MSRP and Price Per Unit have a right-skewed distribution, indicating that while most products are priced in the lower range (45-90 for MSRP, 50-110 for Price Per Unit), there are outliers representing premium-priced products. Although these outliers account for fewer transactions, they drive high-value sales.
- Businesses could consider emphasizing the high-end products in marketing campaigns to capture a premium segment of customers.

Sales and Transaction Size:

- The positive correlation between sales volume and transaction size underscores the fact that products with higher sales are typically associated with larger transactions. This makes it clear that businesses should focus on high-value products, especially those that show consistent sales and larger transactions, to drive revenue growth.

This detailed insight into the data provides a roadmap for tailoring sales strategies, focusing on key regions, product categories, and customer segments, while leveraging trends like seasonality and purchase frequency.

CUSTOMER SEGMENTATION USING RFM ANALYSIS

The primary goal of RFM (Recency, Frequency, Monetary) analysis is to group customers based on their past interactions, allowing businesses to tailor their marketing and sales efforts to the right audience. It helps increase customer retention and maximize revenue by focusing efforts on customers who are most likely to respond.

Number of Customer Segments:

The data will be segmented into four customer groups based on their Recency, Frequency, and Monetary (RFM) values by :

1. Best Customers: Recency = **High**, Frequency = **High**, Monetary = **High**.
2. Loyal Customers: Recency = **Medium** or **Low**, Frequency = **High**, Monetary = **High**.
3. Lost Customers: Recency = **very Low**, Frequency = **very Low**, Monetary = **very Low**.
4. Customers on the Verge of Churning: Recency = **very Low** or **Low**, Frequency = **High or Medium**, Monetary = **very Low or Low**.

This segmentation approach is designed to categorize customers according to their transactional behavior, helping to identify the most valuable customer groups for targeted marketing strategies.

PARAMETERS USED

Recency: Refers to the time elapsed since a customer's last purchase.

- This is calculated by subtracting each customer's most recent order date from the latest order date in the dataset. A shorter time between orders results in a low recency value, placing customers in the "best" category.

Frequency: Measures the total number of purchases made by a customer.

- The frequency parameter is determined by counting how many transactions a customer has completed. Customers with a higher number of transactions are seen as more engaged.

Monetary: Refers to the total amount of money a customer has spent.

- This parameter is based on the total sales attributed to each customer. Customers who spend more are placed into higher monetary bins, considering both recency and frequency of their purchases.

Quartile-Based Binning: To classify customers into these RFM categories based on their RFM values, quartiles (0, 0.25, 0.50, 0.65, 1) are used. This means that each factor (R,F and M) is divided into 4 bins (very low (vL), low (L), medium (M), high(H)), and the bins represent percentiles.

- 0-25% (Low)
- 25-50% (Medium)
- 50-65% (High)
- 65-100% (Very High)

PARAMETERS USED

Recency Bins:

- vL (Very Low) = Oldest orders (more time passed)
- L (Low) = Recent, but not the most recent
- M (Medium) = More recent orders
- H (High) = Most recent orders (least time passed)

A score of 1 in Recency indicates the most recent activity, meaning the customer has made a purchase very recently.

Frequency Bins:

- vL (Very Low) = Few purchases
- L (Low) = Moderate purchases
- M (Medium) = More frequent purchases
- H (High) = Highest number of transactions

A score of 4 for Frequency means the customer has made the highest number of purchases, demonstrating strong engagement with the brand.

Monetary Bins:

- vL (Very Low) = Low spenders
- L (Low) = Moderate spenders
- M (Medium) = Higher spenders
- H (High) = Highest spenders

A score of 4 for Monetary means the customer is a high spender, having contributed the most to the company's revenue.

ASSUMPTIONS FOR CUSTOMER SEGMENTATION:



Best Customers:

- Recency = High: These customers have made recent purchases, indicating they are actively engaged with the business.
- Frequency = High: These customers make purchases frequently, showing they are loyal and consistent in their buying behavior.
- Monetary = High: These customers have spent a significant amount, contributing heavily to revenue.

Assumption:

Best customers are those who are not only active in recent times but also frequently purchase high-value products. These customers are the most valuable to the business, and targeted retention strategies should be focused on them.



Loyal Customers:

- Recency = Medium/Low: These customers have made purchases within a reasonable recent timeframe, but not as frequently as Best Customers.
- Frequency = High: These customers buy regularly, which indicates ongoing interest and engagement.
- Monetary = High: They are high spenders and have contributed significantly to sales over time.

Assumption:

Loyal customers may not have purchased recently but have a strong purchasing history, indicating their ongoing loyalty. Although their recency is not as high as the Best Customers, they still represent a steady source of revenue.

ASSUMPTIONS FOR CUSTOMER SEGMENTATION:



Lost Customers:

- Recency = Very Low: These customers have not made a purchase in a long time, indicating a lack of recent engagement.
- Frequency = Very Low: These customers have made very few or no purchases at all.
- Monetary = Very Low: Their total spending is minimal, making them low-value customers.

Assumption:

Lost customers are those who have become disengaged and are unlikely to return without significant intervention. These customers should be prioritized for re-engagement efforts or considered as "churned."



Customers on the Verge of Churning:

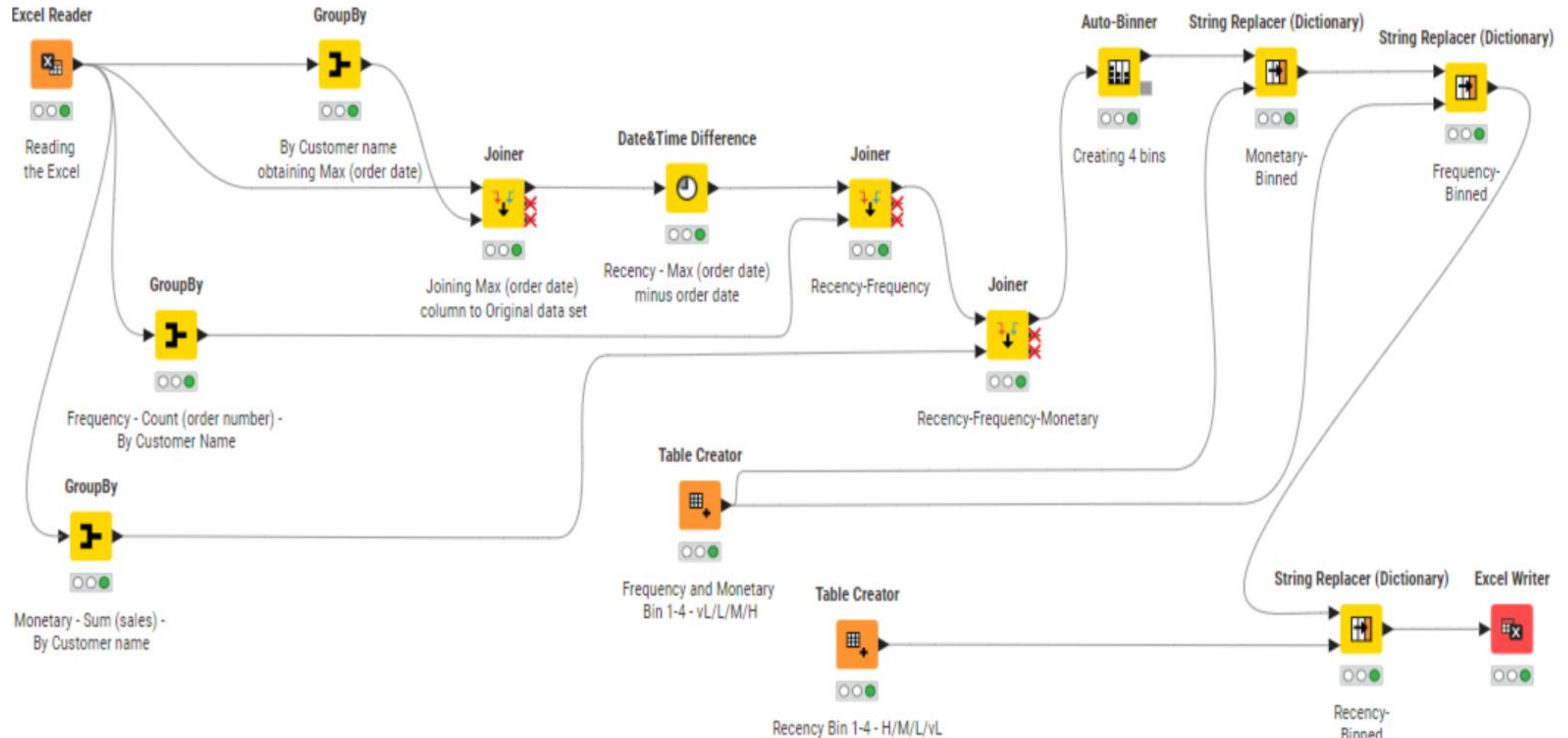
- Recency = Very Low or Low: These customers have not purchased recently or have shown a decrease in purchase frequency.
- Frequency = High or Medium: Despite the recent lack of engagement, these customers have a history of frequent or moderate purchases.
- Monetary = Very Low or Low: Their total spending is lower than expected, indicating a decline in transaction volume.

Assumption:

These customers are at risk of churning. While they were previously active and engaged, their recency and monetary contributions have dropped. These customers could be revived with targeted interventions such as discounts or personalized offers to increase their likelihood of returning.

KNIME WORKFLOW

KNIME is used to perform the RFM Analysis and below is the workflow.



FINAL OUTPUT TABLE

Data Table with updated columns (Table)

Rows: 2747 | Columns: 25

☐	#	R... ↓	ORDERN... Number (inte... ↓	QUANTIT... Number (inte... ↓	PRICEEA... Number (dou... ↓	ORDERLI... Number (inte... ↓	SALES Number (dou... ↓	ORDERD... Local Date ↓	DAYS_SI... Number (inte... ↓	STATUS String ↓	PRODUC... String ↓	MSRP Number (inte... ↓	PRODUC... String ↓	CUSTOM... String ↓	CITY String ↓
☐	1	Row0_	10107	30	95.7	2	2,871	2018-02-24	828	Shipped	Motorcycles	95	S10_1678	Land of Toys In	NYC
☐	2	Row1_	10121	34	81.35	5	2,765.9	2018-05-07	757	Shipped	Motorcycles	95	S10_1678	Reims Collectal	Reims
☐	3	Row2_	10134	41	94.74	2	3,884.34	2018-07-01	703	Shipped	Motorcycles	95	S10_1678	Lyon Souvenir	Paris
☐	4	Row3_	10145	45	83.26	6	3,746.7	2018-08-25	649	Shipped	Motorcycles	95	S10_1678	Toys4GrownUp	Pasadena
☐	5	Row4_	10168	36	96.66	1	3,479.76	2018-10-28	586	Shipped	Motorcycles	95	S10_1678	Technics Store	Burlingame
☐	6	Row5_	10180	29	86.13	9	2,497.77	2018-11-11	573	Shipped	Motorcycles	95	S10_1678	Daedalus Desig	Lille

COUNTRY String ↓	DEALSIZE String ↓	Max(ORD... Local Date ↓	Recency Number (long) ↓	ORDERN... Number (inte... ↓	SALES (S... Number (dou... ↓	Recency [... String ↓	ORDERN... String ↓	SALES (S... String ↓	SALES (S... String ↓	ORDERN... String ↓	Recency [... String ↓
USA	Small	2019-11-15	629	4	164,069.44	Bin 4	Bin 3	Bin 4	H	M	vL
France	Small	2020-03-30	693	5	135,042.94	Bin 4	Bin 4	Bin 3	M	H	vL
France	Medium	2020-03-17	625	3	78,570.34	Bin 4	Bin 1	Bin 1	vL	vL	vL
USA	Medium	2020-01-12	505	3	104,561.96	Bin 4	Bin 1	Bin 2	L	vL	vL
USA	Medium	2020-01-05	434	4	120,783.07	Bin 4	Bin 3	Bin 3	M	M	vL

INFERENCES FROM RFM ANALYSIS



Best Customers

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
ORDERNU	PRODUCTLINE	PRODUCTCODE	CUSTOMERNAME	CITY	COUNTRY	DEALSIZE	Recency	ORDERNUMBER	SALES (Sum)	Recency	ORDERNUMBER	SALES	RECENCY	FREQUENCY	MONETARY	Customer Segment
10347	Classic Cars	S10_1949	Australian Collectors, Co.	Melbourne	Australia	Medium	0	5	200995.41	Bin 1	Bin 4	Bin 4	H	H	H	Best Customer
10424	Classic Cars	S10_1949	Euro Shopping Channel	Madrid	Spain	Large	0	26	912294.11	Bin 1	Bin 4	Bin 4	H	H	H	Best Customer
10406	Classic Cars	S18_1129	Danish Wholesale Imports	Kobenhavn	Denmark	Large	0	5	145041.6	Bin 1	Bin 4	Bin 4	H	H	H	Best Customer
10421	Vintage Cars	S18_2795	Mini Gifts Distributors Ltd.	San Rafael	USA	Medium	0	17	654858.06	Bin 1	Bin 4	Bin 4	H	H	H	Best Customer
10387	Motorcycles	S32_1374	Dragon Souvenirs, Ltd.	Singapore	Singapore	Medium	0	5	172989.68	Bin 1	Bin 4	Bin 4	H	H	H	Best Customer

These customers are the ideal group to focus on. They are recent, frequent, and high-spending, which makes them crucial for your business's immediate revenue growth and long-term sustainability. They're the customers who are most likely to respond well to retention efforts, loyalty programs, and personalized offers. Essentially, they are the golden customers who sustain the business's performance.

The top 5 best customers are performing well across recency, frequency, and monetary values. Classic cars, vintage cars, and motorcycles are the top-selling products, with high order quantities. To maximize revenue, it's important to focus on these product categories, particularly the high-performing product codes, while also giving a boost to the underperforming product codes to drive more sales. Geographically, the countries performing well in the automobile sector should be leveraged further, while additional engagement efforts are needed in the regions showing potential for growth.

INFERENCES FROM RFM ANALYSIS



Loyal Customers

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
ORDERNU	PRODUCTLINE	PRODUCTCODE	CUSTOMERNAME	CITY	COUNTRY	DEALSIZE	Recency	ORDERNUMBER	SALES (Sun	Recency	ORDERNUMBER	SALES (	RECENCY	FREQUENCY	MONETARY	Customer Segment
10223	Motorcycles	S10_1678	Australian Collectors, Co.	Melbourne	Australia	Medium	282	5	200995.4	Bin 3	Bin 4	Bin 4	L	H	H	Loyal Customer
10417	Motorcycles	S10_1678	Euro Shopping Channel	Madrid	Spain	Large	18	26	912294.1	Bin 2	Bin 4	Bin 4	M	H	H	Loyal Customer
10312	Classic Cars	S10_1949	Mini Gifts Distributors Ltd.	San Rafael	USA	Large	221	17	654858.1	Bin 3	Bin 4	Bin 4	L	H	H	Loyal Customer
10277	Classic Cars	S12_4675	Dragon Souvenirs, Ltd.	Singapore	Singapore	Medium	211	5	172989.7	Bin 3	Bin 4	Bin 4	L	H	H	Loyal Customer
10256	Vintage Cars	S18_1342	Danish Wholesale Imports	Kobenhavn	Denmark	Medium	312	5	145041.6	Bin 3	Bin 4	Bin 4	L	H	H	Loyal Customer

Loyal customers are important because they represent sustained engagement and steady revenue. While they might not have purchased as recently or frequently as Best Customers, they still have high potential for future purchases.

They may be at a point where their activity has decreased slightly, but with the right interventions, they can return to being top customers. Loyal customers are often high-value and brand advocates, so it's crucial to focus on maintaining their loyalty, for example, through loyalty programs or personalized re-engagement strategies.

The top 5 loyal customers are excelling in terms of frequency and monetary value, with motorcycles leading the way. While these customers are performing well, a slight push could elevate them to the Best Customers category, particularly by improving their recency. To drive further revenue, the focus should be on these high-performing product categories, especially the top-selling product codes.

INFERENCES FROM RFM ANALYSIS



Lost Customers

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
ORDERNU	PRODUCTLINE	PRODUCTCODE	CUSTOMERNAME	CITY	COUNTRY	DEALSIZE	Recency	ORDERNU	SALES (Sur	Recency	ORDERNUMBER	SALES	RECENCY_	FREQUEN	MONETAR	Customer Segment
10124	Vintage Cars	S18_1749	Signal Gift Stores	Las Vegas	USA	Small	557	3	82751.08	Bin 4	Bin 1	Bin 1	vL	vL	vL	Lost Customer
10147	Classic Cars	S12_1099	Collectables For Less Inc.	Brickhaver	USA	Large	502	3	81577.98	Bin 4	Bin 1	Bin 1	vL	vL	vL	Lost Customer
10241	Classic Cars	S18_1589	Mini Caravy	Strasbourg	France	Small	367	3	80438.48	Bin 4	Bin 1	Bin 1	vL	vL	vL	Lost Customer
10134	Motorcycles	S10_1678	Lyon Souvenirs	Paris	France	Medium	625	3	78570.34	Bin 4	Bin 1	Bin 1	vL	vL	vL	Lost Customer
10118	Ships	S700_3505	Enaco Distributors	Barcelona	Spain	Medium	582	3	78411.86	Bin 4	Bin 1	Bin 1	vL	vL	vL	Lost Customer

Lost customers are those who have become disengaged and have stopped generating value for the business. Their lack of recent activity, low frequency, and minimal spending all indicate that they are no longer interacting with the brand. However, if they were valuable in the past, there may still be an opportunity to re-engage them through targeted marketing efforts.

The top 5 lost customers are from the USA, France, and Spain, regions where sales have been strong, and performance is good. Interestingly, these lost customers are associated with classic cars, vintage cars, and motorcycles, which are top-selling categories. Further data is needed to understand the reasons behind their disengagement, as this will be crucial in determining whether targeted marketing strategies can effectively bring them back.

INFERENCES FROM RFM ANALYSIS



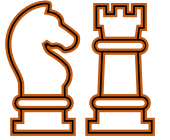
Customers on the verge of churning

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
ORDERNU	PRODUCTLINE	PRODUCTCODE	CUSTOMERNAME	CITY	COUNTRY	DEALSIZE	Recency	ORDERNU	SALES (Sur	Recency [t	ORDERNU	SALES	RECENCY	FREQUENCY	MONETARY	Customer Segment
10103	Classic Cars	S10_1949	Baane Mini Imports	Stavern	Norway	Medium	645	4	116599.2	Bin 4	Bin 3	Bin 2	vL	M	L	Customer on the Verge of Churning
10112	Classic Cars	S10_1949	Volvo Model Replicas, Co	Lule	Sweden	Large	605	4	75754.88	Bin 4	Bin 3	Bin 1	vL	M	vL	Customer on the Verge of Churning
10217	Classic Cars	S10_4962	Handji Gifts& Co	Singapore	Singapore	Large	444	4	115498.7	Bin 4	Bin 3	Bin 2	vL	M	L	Customer on the Verge of Churning
10300	Classic Cars	S12_1099	Blauer See Auto, Co.	Frankfurt	Germany	Medium	397	4	85171.59	Bin 4	Bin 3	Bin 2	vL	M	L	Customer on the Verge of Churning
10116	Trains	S32_3207	Rovale Belge	Charleroi	Belgium	Small	639	4	33440.1	Bin 4	Bin 3	Bin 1	vL	M	vL	Customer on the Verge of Churning

These customers are showing signs of declining engagement and are at risk of churning. While they are still somewhat engaged (based on frequency), their low recency and low monetary value indicate a drop in interest and spending. These customers are still valuable because they have a history of frequent purchases or moderate activity. This group requires urgent action to prevent them from fully churning.

The top 5 customers listed below show a high monetary contribution, with classic cars being the primary category that requires further promotion. Given the large sales, there is a need to focus on boosting medium and large sales category products to engage these customers. Specific product codes should be given extra attention to drive more interest. These customers need just a small push to increase their purchase frequency, and taking immediate action could elevate them to a higher tier.

TARGETED MARKETING STRATEGIES



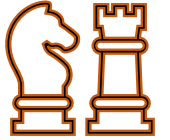
1. Best Customers

- To strengthen the bond with your top customers, offer VIP or loyalty rewards, granting access to exclusive discounts, early product releases, or members-only events.
- USA, Spain, France and Australia could be targeted more.
- Create personalized offers using their purchase history and preferences, providing tailored product recommendations or exclusive deals. For example, offer direct promotions for high-performing items like S18_3232 and S10_1949 in the Classic Cars category.
- Leverage seasonal peaks (especially Q4) by giving Best Customers early access to premium collections like Classic Cars or Vintage Cars, targeting high-demand seasons such as October-November.
- Introduce a refer-a-friend program with discounts on premium products like Classic Cars, rewarding customers who successfully refer new buyers.

2. Loyal Customers

- Send targeted email offers to loyal customers with medium recency, offering time-sensitive discounts on high-demand items like Motorcycles and Vintage Cars.
- With the country with highest sales, the countries having high revenue generating customers from Austria and Singapore could be targeted.
- Highlight items that are back in stock or new arrivals in their preferred categories.
- Since these customers have medium recency, remind them of products they previously purchased or showed interest in, reigniting their engagement.
- Based on their frequent purchases and high spending, offer exclusive bundle deals, such as discounted Motorcycle parts or bundled packages combining high-value products like Motorcycles and Classic Cars.

TARGETED MARKETING STRATEGIES



3. Customers on the Verge of Churning

- Launch a win-back email campaign offering personalized discounts on products they've previously purchased or similar high-value items like Classic Cars or Motorcycles.
- Tailor offers based on their past purchase behavior or preferences, sending emails with a special offer or loyalty bonus to encourage their return.
- Create a sense of urgency with limited-time discounts or special deals, using FOMO (Fear of Missing Out) tactics to prompt them to make a purchase.
- Using seasonality to promote the products by giving additional discounts specifically to the identified customers would generate positive response.

4. Lost Customers

- Send a re-engagement offer with an irresistible discount or free shipping for returning customers.
- For high-value lost customers, consider direct outreach (phone call or personalized email) to ask why they stopped buying and offer a tailored incentive for their return.
- Use retargeting ads on social media or Google to re-engage lost customers with seasonal campaigns or promotions that may appeal to them based on past behaviours.
- As the top customers are from the regions with high sales, reaching out to them personally would help knowing the reason for the fall out.

These campaigns will be more effective because they are based on insights from customer behavior, preferences, geographic distribution, and seasonality, leveraging the specific variables identified in the EDA.